Tedtronics — perfboard revision 5

7×10 cm stock · two banks · six interchangeable positions · three active capsules

ENGINEERING PROTOTYPE — NOT RELEASED FOR SOLDERING

Software checks: **pass**. Actual hardware, module envelopes, timing and electrical performance remain pending. Read the release gates before soldering.

Architecture

- ESP1 owns routing, rail enable and fault reset.
- ESP2 owns smart modes, voltage scanning and joystick ADC1 inputs.
- Each bank shares an ADS1115 voltage scanner and buffered selector.
- Every position retains independent output clamps and series-path isolation.
- Pi and its peripherals have a separate supply. RC522 uses the purchased MIFARE tags.
- Socket through-hole ICs. Carrier interfaces are labelled pigtails; their connector numbering is not IC numbering.

Construction comparison

Metric	Legacy	Revision 5
Boards	17	27
Functional parts	551	457
Adapter assemblies	47	23
Local wire segments	1511	1392
Board solder estimate	5224	4946

The legacy design used larger boards. Revision 5 also has 341 distribution posts and 279 harness conductors. Counts exclude carrier internals and adapter/socket preparation. This revision reduces adapters; it is not a claim of fewer boards or a finished minimum-work optimum.

Electrical bounds and release gates

Calculated design bounds

- Voltage input impedance: ≥1.98 MΩ with 1% divider resistors.
- Worst divided input at 5.25 V: 2.919 V.
- Static source/sink clamp maximum: 11.08 mA.
- Current shunt: 1 Ω ±1%; 10 mV shunt burden at 10 mA, plus switch and wiring resistance.
- Bank scan target: ≤20 ms. Wait for conversion-ready, discard after switching and keep timestamps.
- Watchdog expiry target: 200–350 ms; actual HC123 timing must be measured before permanent installation.
- 5 V guard static trip: 5.053–5.227 V. Approximately 23 mV margin below the contact ceiling does not establish safe fault transients.

These calculations do not establish loop stability, transient limits or measured accuracy.

Pending physical evidence

- Actual stock grid and plating
- Pogo pin diameter, compressed height and mounting fit
- DevKit pinout and body measurements
- Chosen ADS1115/INA226 carrier interfaces and envelopes
- Bench qualification of output clamps, supply guards and partial-power states
- Firmware implementation and measured acceptance

Verify the exact carrier pads, actual stock grid, socket bodies, compressed pogo heights and inter-board mounting offsets.

Current contact geometry needs **156 pogo pins**. The inventory lists 150, leaving a six-pin shortage before spares.

The 12 V adapter provides a nominal 24 W budget. Measure fan, light, controller and buck input loads. Do not include the Pi in this budget.

CP00 — First light

Filmable win: ESP32 switches the real four-column LED loop.

Installation

Add 19 components/posts, 46 local wires and 0 harness conductors. Earlier permanent work stays in place. Boards receiving parts: a1, router-core. • Prototype source only: positive C2 / ground C1; final supply and matrix disconnected.
• OLED uses ESP1 GPIO4 SDA / GPIO16 SCL at 3.3V. Remove diagnostic probes before ordinary operation.

Firmware handoff

factory LED: GPIO32 defaults LOW; explicit enable/disable; GPIO25 stimulus LOW; all future routing/output controls LOW; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Identify the exact classic ESP32 DevKit pinout, the USB-powered 5V output and 3V3 output. Identify actual pogo and adapter dimensions before soldering. Confirm the four kit pieces fit the selected rows.
• Unpowered A1: all six contacts within each column <1Ω; different columns >10MΩ with external links isolated. Inspect pin1 and the NODE header mapping.
• Prototype disconnected from A1: measure USB 5V ≤5.25V and 3V3 3.20–3.38V. R_LIMIT is 560Ω ±1%, ≥0.25W. With its output shorted, source current ≤9.5mA; remove short before connecting the kit.
• Place fake battery negative C1 / positive C2, 1kΩ resistor C2–C3, LED anode C3 / cathode C4, and kit jumper C4–C1. Prototype J_OUTPUT pin1→A1 J_NODE pin3 and pin2→pin1. No final-power or matrix cable connected.
• Compatible factory firmware: GPIO32 LOW/reset makes LED OFF; explicit enable lights LED; disable turns it OFF. Record USB-test-source voltage at C2 under load; it is intentionally unregulated and must not be called the final 5V rail.

Film the result

Show first light setup with power OFF and display firmware revision. Demonstrate: ESP32 switches the real four-column LED loop. Show the documented negative test and actual recorded readings.
Negative example: Disable or reset the ESP32: the LED must go dark without touching the kit.
Record actual readings, equipment, board/firmware revision and link positions.

CP01 — Reject an open contact

Filmable win: A lifted jumper contact is electronically rejected.

Installation

Add 38 components/posts, 66 local wires and 2 harness conductors. Earlier permanent work stays in place. Boards receiving parts: measurement-a-precision. • USB test-source positive lead unplugged; final supply and matrix leads absent.
• TEST_A→A1 C4, TEST_B→C1; JP_RETURN fitted only during diagnostic test.
• Before LED enable: stimulus OFF, JP_RETURN/probes removed, prototype positive restored with power OFF.
• Use bank-A precision input pair for TEST_A=C4 / TEST_B=C1 through removable sense leads. Its permanent NODE00/NODE01 harness is absent until CP06. Select CD4052 channel0; verify actual ADC/buffer supply before stimulus.

Firmware handoff

factory fixed-pair continuity: GPIO4 SDA,16 SCL,25 stimulus; ADS1115 0x48; manual source isolation; reject open/wrong/inconclusive, fresh test + explicit enable; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Student supply OFF and positive fixed-battery lead unplugged. TEST_A→C4, TEST_B→C1; fit JP_RETURN only during test. Check stimulus current ≤0.34mA and educational contact voltages 0–5.25V.
• Calibrate open, a known short and 1kΩ. Measure open stimulus voltage Vs with DMM; infer R=11k×V/(Vs−V), accounting for divider gain and input loading. Known short ≤20Ω; ≥100Ω rejected; 20–100Ω or saturated/unstable readings inconclusive and OFF. Compare inferred short with DMM continuity <1Ω.
• Test the actual four-piece scene: correct C4–C1 kit jumper passes; one lifted contact and wrong-node placements fail. Include alternate paths through LED/resistor in calibration; no false pass through a component.

Film the result

Show reject an open contact setup with power OFF and display firmware revision.
Demonstrate: A lifted jumper contact is electronically rejected. Show the documented negative test and actual recorded readings.
Negative example: Lift the return jumper while retaining its convincing visual position: the electronic test rejects it.
Record actual readings, equipment, board/firmware revision and link positions.

CP02 — Reject a wrong node

Filmable win: A convincing-looking jumper on the wrong node is rejected.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • USB test-source positive lead unplugged; final supply and matrix leads absent.

• TEST_A→A1 C4, TEST_B→C1; JP_RETURN fitted only during diagnostic test.

• Before LED enable: stimulus OFF, JP_RETURN/probes removed, prototype positive restored with power OFF.

• Use bank-A precision input pair for TEST_A=C4 / TEST_B=C1 through removable sense leads. Its permanent NODE00/NODE01 harness is absent until CP06. Select CD4052 channel0; verify actual ADC/buffer supply before stimulus.

Firmware handoff

factory fixed-pair continuity: GPIO4 SDA,16 SCL,25 stimulus; ADS1115 0x48; manual source isolation; reject open/wrong/inconclusive, fresh test + explicit enable; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Test the actual four-piece scene: correct C4–C1 kit jumper passes; one lifted contact and wrong-node placements fail. Include alternate paths through LED/ resistor in calibration; no false pass through a component.

• Simulated camera approval + failed/inconclusive electrical result keeps student supply OFF. Correction requires a fresh electrical test and separate enable request.

Film the result

Show reject a wrong node setup with power OFF and display firmware revision.
Demonstrate: A convincing-looking jumper on the wrong node is rejected. Show the documented negative test and actual recorded readings.
Negative example: Connect the jumper to a different column: electronic rejection persists despite simulated camera approval.
Record actual readings, equipment, board/firmware revision and link positions.

CP03 — Correction needs a fresh test

Filmable win: Correcting the jumper requires a fresh pass and a separate enable request.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • USB test-source positive lead unplugged; final supply and matrix leads absent.

• TEST_A→A1 C4, TEST_B→C1; JP_RETURN fitted only during diagnostic test.

• Before LED enable: stimulus OFF, JP_RETURN/probes removed, prototype positive restored with power OFF.

• Use bank-A precision input pair for TEST_A=C4 / TEST_B=C1 through removable sense leads. Its permanent NODE00/NODE01 harness is absent until CP06. Select CD4052 channel0; verify actual ADC/buffer supply before stimulus.

Firmware handoff

factory fixed-pair continuity: GPIO4 SDA,16 SCL,25 stimulus; ADS1115 0x48; manual source isolation; reject open/wrong/inconclusive, fresh test + explicit enable; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Before enable: stimulus OFF, JP_RETURN removed, test leads disconnected, fixed battery lead restored with power off. No automatic enable after a passing test.

• Simulated camera approval + failed/inconclusive electrical result keeps student supply OFF. Correction requires a fresh electrical test and separate enable request.

Film the result

Show correction needs a fresh test setup with power OFF and display firmware revision. Demonstrate: Correcting the jumper requires a fresh pass and a separate enable request. Show the documented negative test and actual recorded readings.

Negative example: Refit the correct jumper without a new test: it must remain OFF until a fresh pass and separate enable request.

Record actual readings, equipment, board/firmware revision and link positions.

CP04 — Capsule greeting

Filmable win: One capsule displays a real pt-BR greeting.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • Prototype source only: positive C2 / ground C1; final supply and matrix disconnected.

• OLED uses ESP1 GPIO4 SDA / GPIO16 SCL at 3.3V. Remove diagnostic probes before ordinary operation.

Firmware handoff

factory OLED on ESP1: GPIO4 SDA /16 SCL, 0x3C; report identity and OFF/ready; no relay requests; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Connect only one OLED at 0x3C to ESP1 FIX_SDA GPIO4 / FIX_SCL GPIO16 using 3.3V logic and removable leads. Use one effective set of pull-ups; continuity fixture must be unplugged when it is not used. Confirm carrier pin labels before power.

• OLED displays greeting with educational source OFF. Display unplug/reconnect and reset do not automatically enable the source.

Film the result

Show capsule greeting setup with power OFF and display firmware revision.

Demonstrate: One capsule displays a real pt-BR greeting. Show the documented negative test and actual recorded readings.

Negative example: Unplug the OLED with power OFF: report missing hardware; a simulated greeting is not a pass.

Record actual readings, equipment, board/firmware revision and link positions.

CP05 — Identity and OFF status

Filmable win: The capsule displays its identity and actual OFF/ready state.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • Prototype source only: positive C2 / ground C1; final supply and matrix disconnected.

• OLED uses ESP1 GPIO4 SDA / GPIO16 SCL at 3.3V. Remove diagnostic probes before ordinary operation.

Firmware handoff

factory OLED on ESP1: GPIO4 SDA /16 SCL, 0x3C; report identity and OFF/ready; no relay requests; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Display position A1, firmware revision, and Desligado/Pronto states. Never substitute a simulated GUI status for an actual hardware observation.

Film the result

Show identity and off status setup with power OFF and display firmware revision.

Demonstrate: The capsule displays its identity and actual OFF/ready state. Show the documented negative test and actual recorded readings.

Negative example: Change identity or remove the capsule: retain OFF and require a new configuration.

Record actual readings, equipment, board/firmware revision and link positions.

CP06 — Voltage calibration

Filmable win: The voltage input matches known zero and reference readings.

Installation

Add 14 components/posts, 25 local wires and 10 harness conductors. Earlier permanent work stays in place. Boards receiving parts: a1, bank-a-position-0. • Prototype source only: positive C2 / ground C1; final supply and matrix disconnected.

• OLED uses ESP1 GPIO4 SDA / GPIO16 SCL at 3.3V. Remove diagnostic probes before ordinary operation.

• Reuse bank-A ADC/buffer. Remove diagnostic probes, JP_RETURN and stimulus leads; attach permanent NODE00/NODE01 harness. Connect service position0 after inspecting pull-ups; no duplicate ADC fixture.

Firmware handoff

factory voltage: ESP1 OLED + local ADS1115 0x48; USB3V3 to isolated card supply via removable links; P3/ammeter disconnected; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Verify ADS1115 0x48. Sweep P1/P2 within 0–5.25V; compare displayed signed difference with DMM at 0,1,2.5,5V. Error ≤0.10V+5%|reading|, impedance ≥1MΩ, update rate ≥10Hz.

• Reverse sense polarity to show a negative difference; do not apply negative absolute contact voltages. P3 remains disconnected and ammeter paths open.

Film the result

Show voltage calibration setup with power OFF and display firmware revision.
Demonstrate: The voltage input matches known zero and reference readings. Show the documented negative test and actual recorded readings.
Negative example: Apply a known reference outside the accepted calibration tolerance: reject the calibration.
Record actual readings, equipment, board/firmware revision and link positions.

CP07 — Read real voltage

Filmable win: The capsule reading follows the multimeter as loop voltage changes.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • Prototype source only: positive C2 / ground C1; final supply and matrix disconnected.
• OLED uses ESP1 GPIO4 SDA / GPIO16 SCL at 3.3V. Remove diagnostic probes before ordinary operation.

Firmware handoff

factory voltage: ESP1 OLED + local ADS1115 0x48; USB3V3 to isolated card supply via removable links; P3/ammeter disconnected; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• With prototype source and diagnostic probes isolated as specified, compare capsule V(P1)−V(P2) with DMM at zero and the actual limited-source voltage. Error ≤0.10V+5%|reading|; ≥10 updates/s. All node absolute voltages 0–5.25V.

Film the result

Show read real voltage setup with power OFF and display firmware revision.
Demonstrate: The capsule reading follows the multimeter as loop voltage changes.
Show the documented negative test and actual recorded readings.
Negative example: Remove the measured source: display the actual near-zero voltage, not a cached reading.
Record actual readings, equipment, board/firmware revision and link positions.

CP08 — Signed voltage

Filmable win: Swapping only the measurement leads reverses the reading sign.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • Prototype source only: positive C2 / ground C1; final supply and matrix disconnected.
• OLED uses ESP1 GPIO4 SDA / GPIO16 SCL at 3.3V. Remove diagnostic probes before ordinary operation.

Firmware handoff

factory voltage: ESP1 OLED + local ADS1115 0x48; USB3V3 to isolated card supply via removable links; P3/ammeter disconnected; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Power OFF before swapping P1/P2 removable measurement leads. Compare signed differential reading with DMM in both orientations; never apply a negative absolute node voltage. P3 and ammeter paths remain disconnected.

Film the result

Show signed voltage setup with power OFF and display firmware revision.
Demonstrate: Swapping only the measurement leads reverses the reading sign.
Show the documented negative test and actual recorded readings.
Negative example: Reverse the test leads: the sign must reverse with the same magnitude tolerance.
Record actual readings, equipment, board/firmware revision and link positions.

CP01 — removable measurement leads

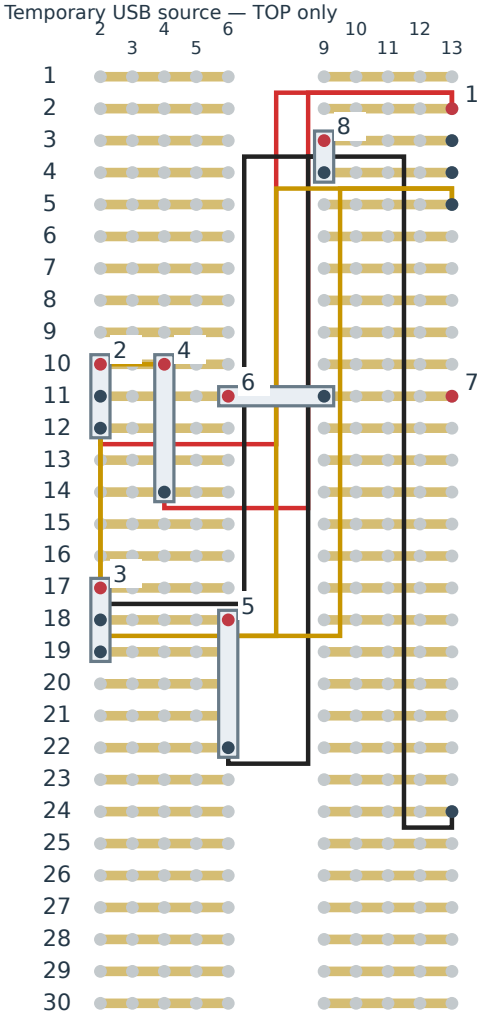
Student source positive unplugged. No matrix. Keep these leads distinct from permanent harnesses.

Purpose	From	To / condition
Temporary precision input A	measurement-a-precision / J_LINK_4.1	A1 C4 / NODE03; also J_DIAGNOSTIC.1 stimulus
Temporary precision input B	measurement-a-precision / J_LINK_5.1	A1 C1 / NODE00; also J_DIAGNOSTIC.2 return
Temporary 3.3 V supply	router-core J_FIXTURE.1	measurement-a-precision / J_LINK_3.1
Common return	router-core J_FIXTURE.2	measurement-a-precision JP_RETURN.2 / GND; permanent ground harness may supply this
Temporary ADC SDA	router-core J_FIXTURE.3 / GPIO4	measurement-a-precision J_A_ADC_MODULE.4 / SDA
Temporary ADC SCL	router-core J_FIXTURE.4 / GPIO16	measurement-a-precision J_A_ADC_MODULE.3 / SCL
Stimulus control	router-core J_FIXTURE.5 / GPIO25	measurement-a-precision / J_LINK_6.1; use the CP01 permanent harness
Test return link	measurement-a-precision JP_RETURN.1	JP_RETURN.2; fit only during unpowered continuity test

The input terminals retain their final net names even while temporary leads test C4/C1. Their permanent NODE00/NODE01 harness is absent until CP06. CD4052 selector pull-downs choose channel0; leave later ESP2 selector harnesses disconnected. At CP06 remove temporary sense/stimulus leads and JP_RETURN before installing the permanent node harness. The USB3V3 feed is removed before final regulated-rail links are fitted at CP11; temporary ESP1 bus control ends at CP15.

CP00 — temporary USB source

Solderless fixture; excluded from the 7×10 cm permanent-board inventory



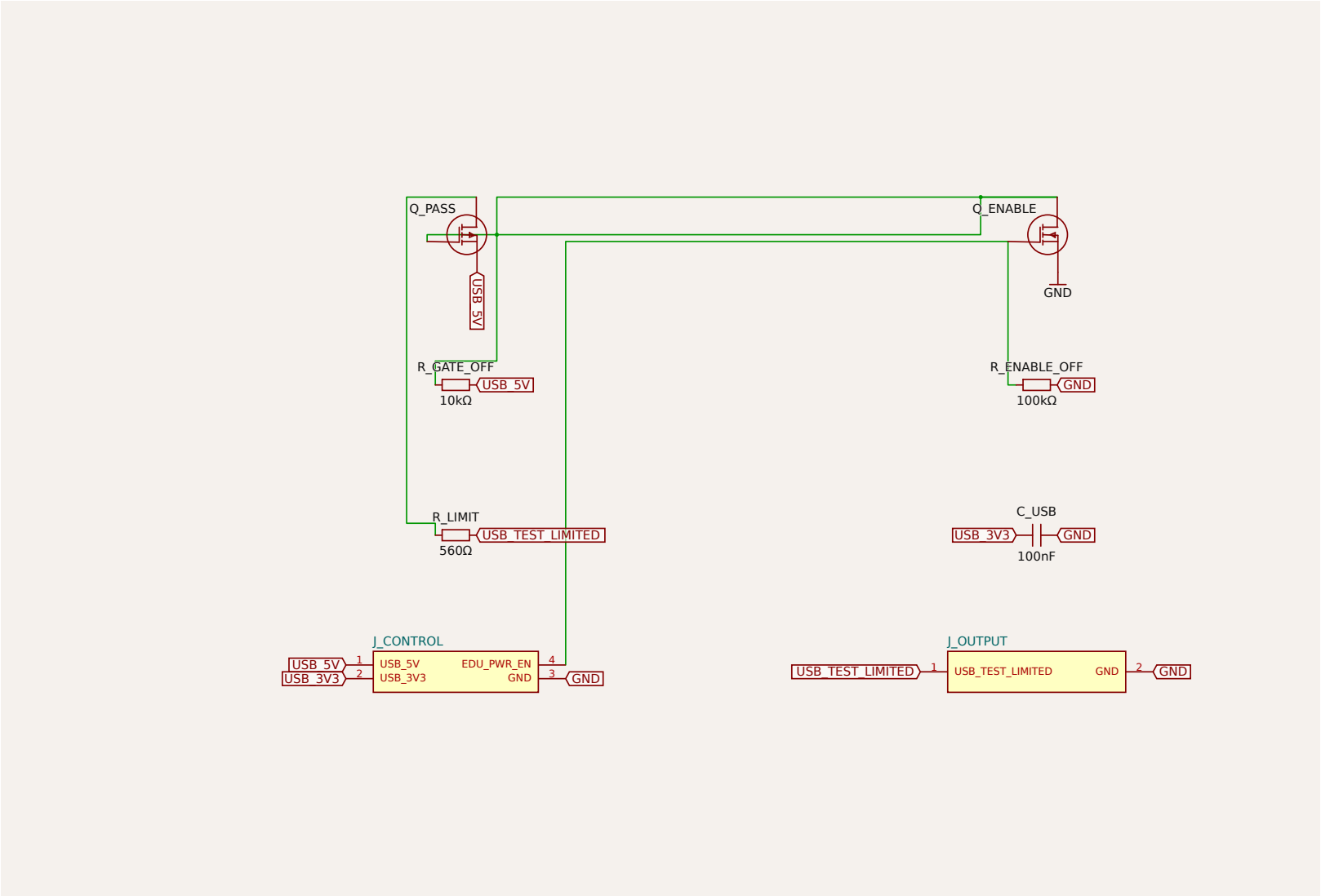
Gold: factory five-hole row ties. Check your fixture.

Coordinates follow this drawing; no underside soldering.

# / reference	Purpose
1 / J_CONTROL	Four individual flying leads from ESP1 prototype header
2 / Q_PASS	USB high-side switch
3 / Q_ENABLE	3.3V gate-control transistor
4 / R_GATE_OFF	10kΩ makes the P-channel switch OFF
5 / R_ENABLE_OFF	100kΩ holds enable LOW
6 / R_LIMIT	560Ω ±1%, 0.25W; bounds every educational-source path
7 / J_OUTPUT	Two flying leads: C2 positive / C1 return
8 / C_USB	USB3V3 fixture decoupling

J_CONTROL pins 1…4 receive router J_PROTOTYPE pins 1…4: USB5V, USB3V3, GND, GPIO32 enable. Q_PASS is NDP6020P (G/D/S); Q_ENABLE is 2N7000 (S/G/D). Confirm the purchased lead order. The five-hole groups are factory-connected; do not add wires for the listed ties.

CP00 — temporary-source schematic



CP00 — fixture jumpers and kit loop

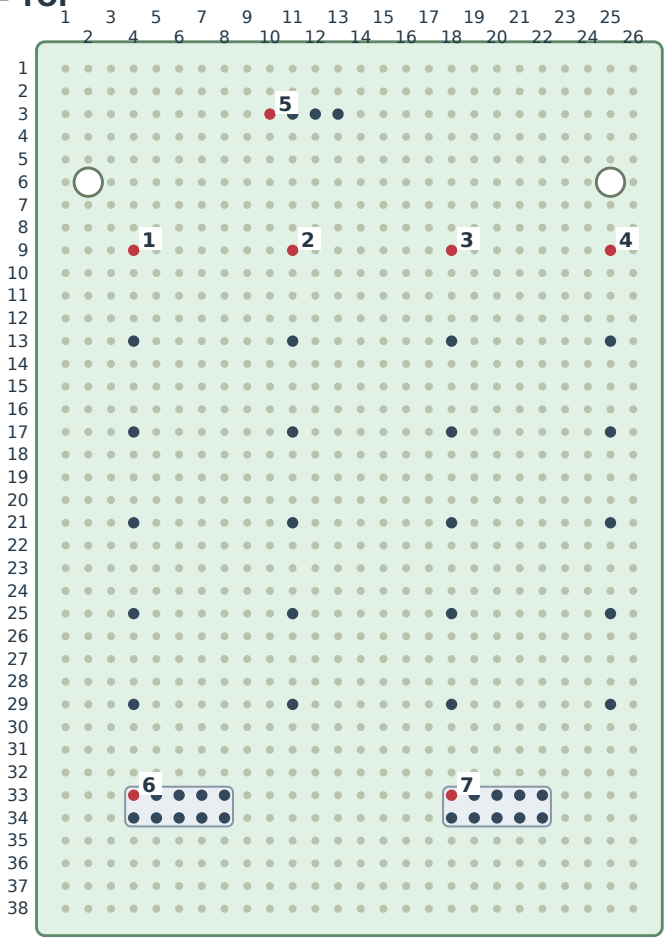
Wire / net	Endpoints	TOP holes
W1 / USB_5V	J_CONTROL.1 → Q_PASS.3	13,2 → 2,12
W2 / USB_5V	J_CONTROL.1 → R_GATE_OFF.2	13,2 → 4,14
W5 / GND	C_USB.2 → J_OUTPUT.2	9,4 → 13,24
W6 / GND	C_USB.2 → Q_ENABLE.1	9,4 → 2,17
W7 / GND	C_USB.2 → R_ENABLE_OFF.2	9,4 → 6,22
W8 / EDU_PWR_EN	J_CONTROL.4 → Q_ENABLE.2	13,5 → 2,18
W9 / EDU_PWR_EN	J_CONTROL.4 → R_ENABLE_OFF.1	13,5 → 6,18
W10 / PROTO_GATE	Q_ENABLE.3 → Q_PASS.1	2,19 → 2,10
W11 / PROTO_GATE	Q_ENABLE.3 → R_GATE_OFF.1	2,19 → 4,10

Factory ties — no added jumper C_USB.1 → J_CONTROL.2 (USB_3V3)
C_USB.2 → J_CONTROL.3 (GND)
Q_PASS.2 → R_LIMIT.1 (PROTO_DRAIN)
J_OUTPUT.1 → R_LIMIT.2 (USB_TEST_LIMITED)
Four top-side kit pieces Fake battery: C1 negative / C2 positive.
1 kΩ resistor: C2 → C3.
LED: C3 anode → C4 cathode.
Jumper: C4 → C1.
Source output pin1 → A1 J_NODE pin3 (NODE01 / C2); output pin2 → pin1 (NODE00 / C1).
No protected-power board or matrix is required. Verify reset/disable extinguishes the LED and the disconnected shorted source stays ≤9.5 mA before attaching the kit.

a1

A1 contacts

a1 — TOP

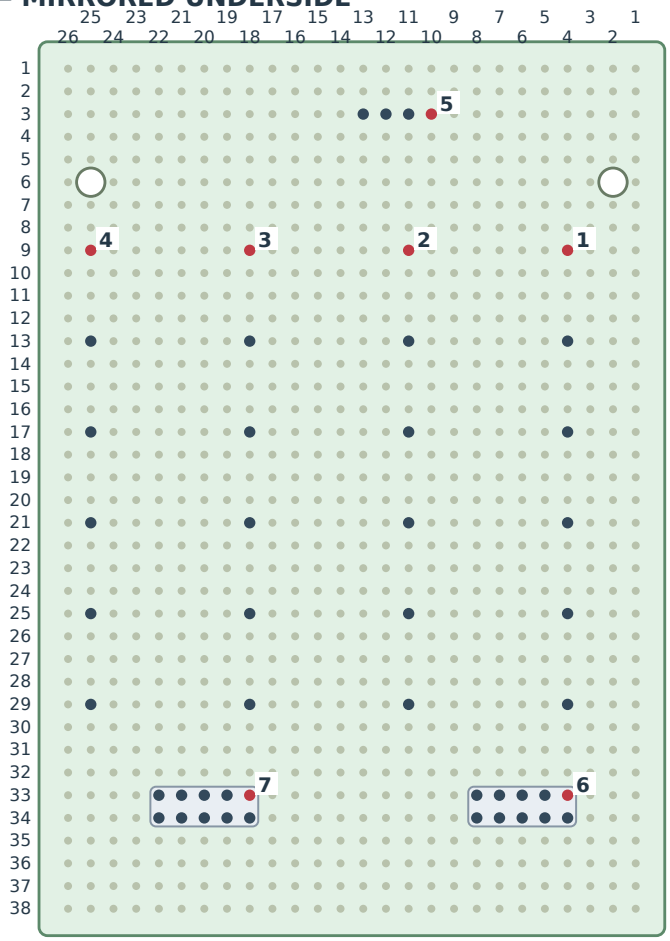


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

a1 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

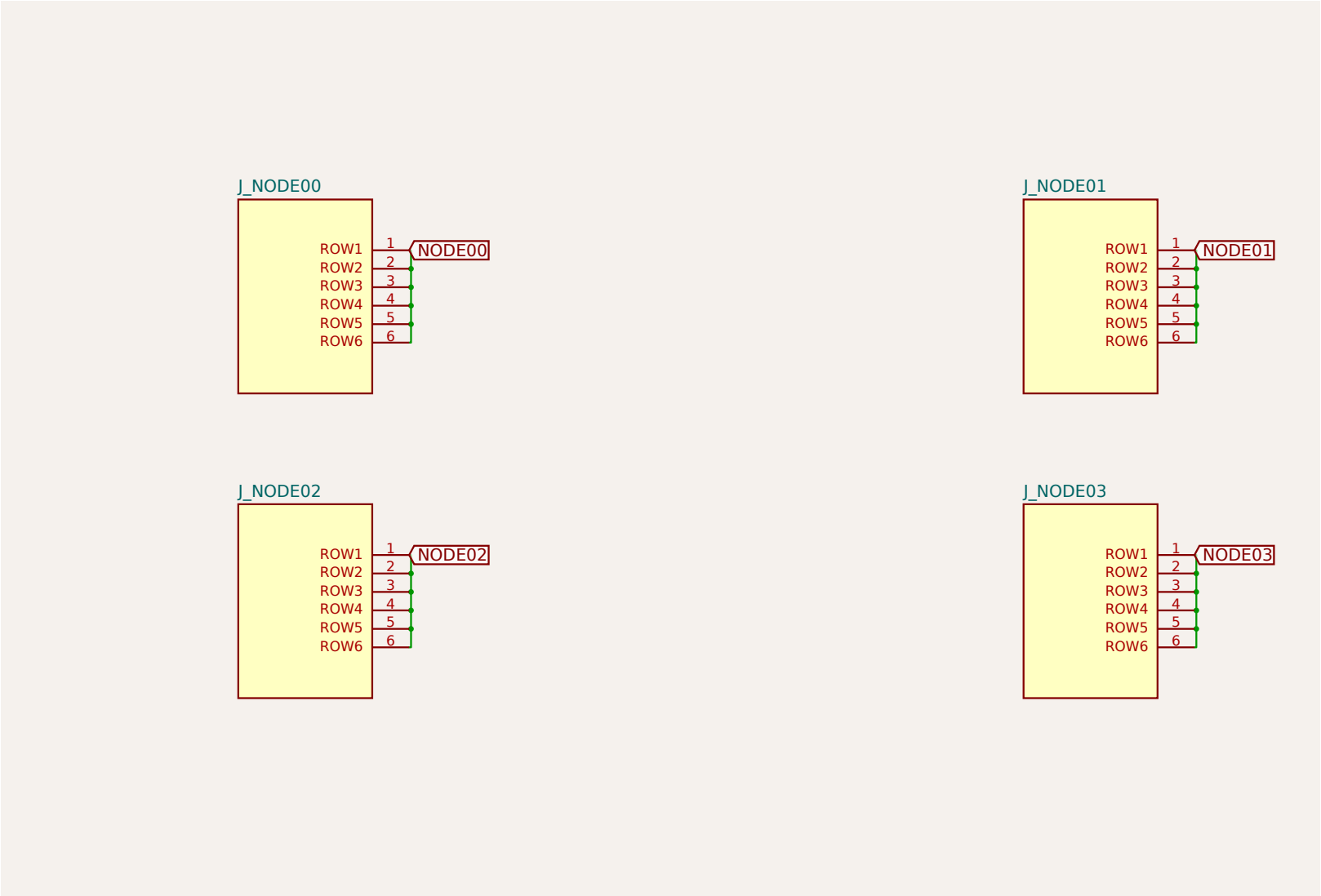
PROTOTYPE: physical fit and hardware tests pending.

a1 — numbered parts

Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

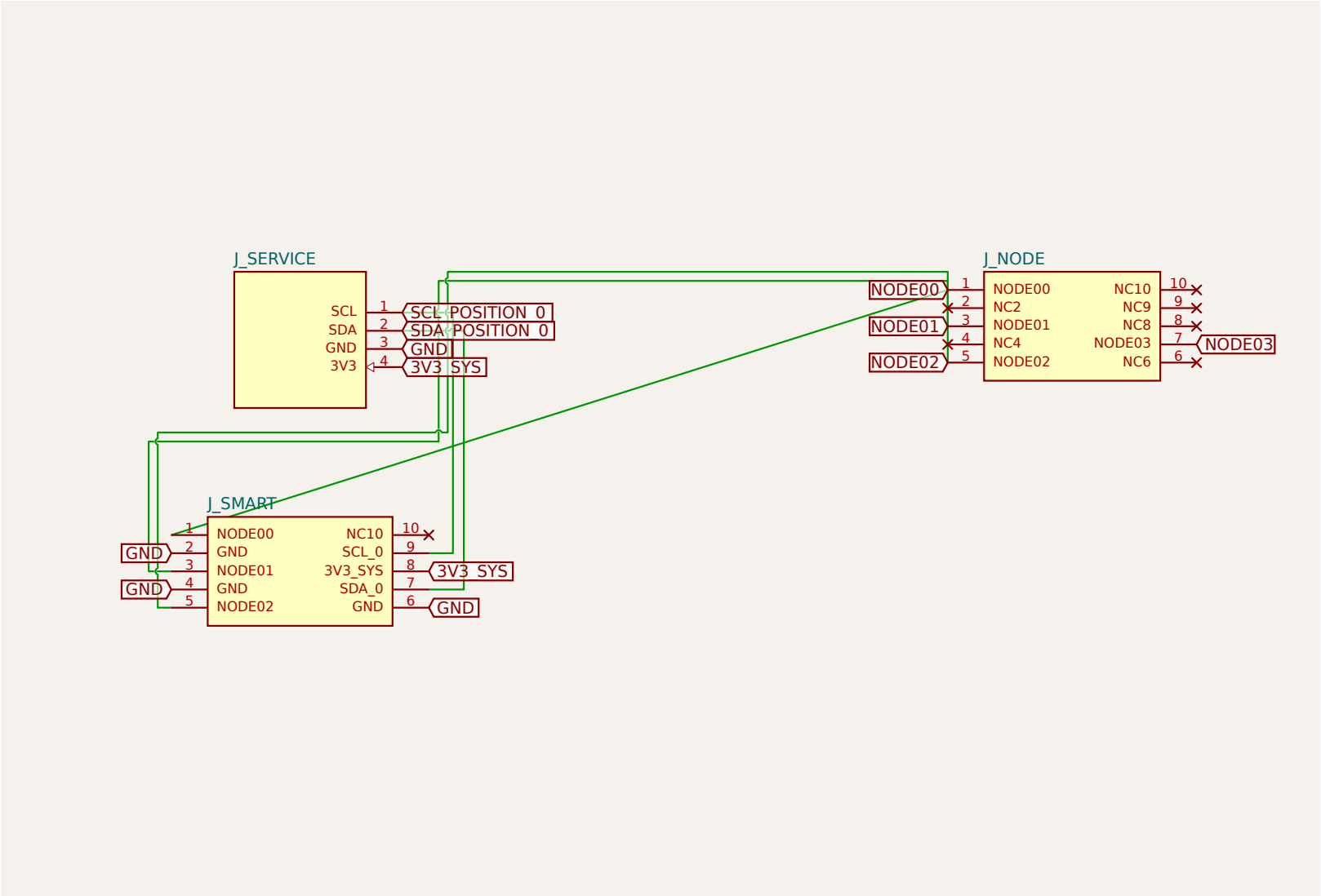
#	Reference	Purpose	Install	Adapter / carrier
1	J_NODE00	NODE00 — six contacts	CP0	
2	J_NODE01	NODE01 — six contacts	CP0	
3	J_NODE02	NODE02 — six contacts	CP0	
4	J_NODE03	NODE03 — six contacts	CP0	
5	J_SERVICE	OLED service pogo contacts	CP6	
6	J_NODE	NODE cable → matrix	CP0	
7	J_SMART	SMART_RAW v2 → channel 0	CP6	

a1 — Nodes



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

a1 — Service

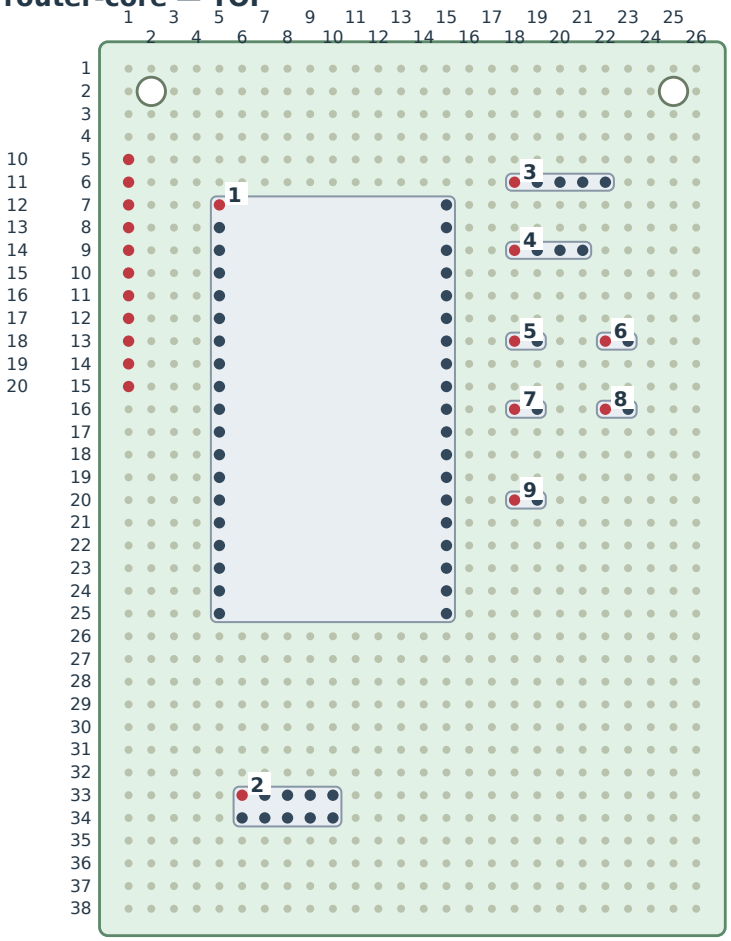


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

router-core

ESP1 carrier and enclosure I/O

router-core — TOP

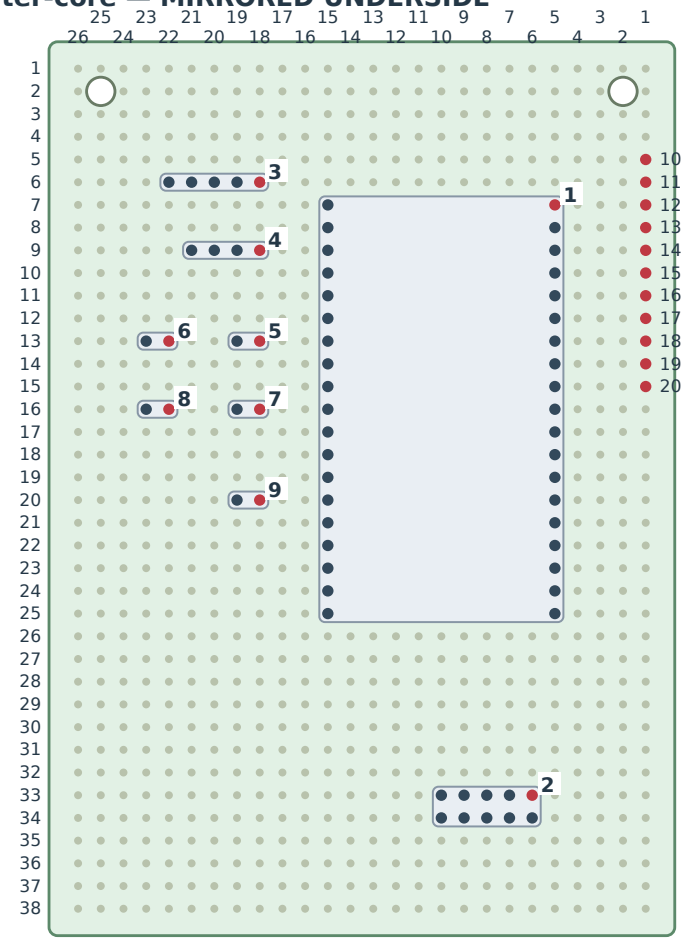


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

router-core — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

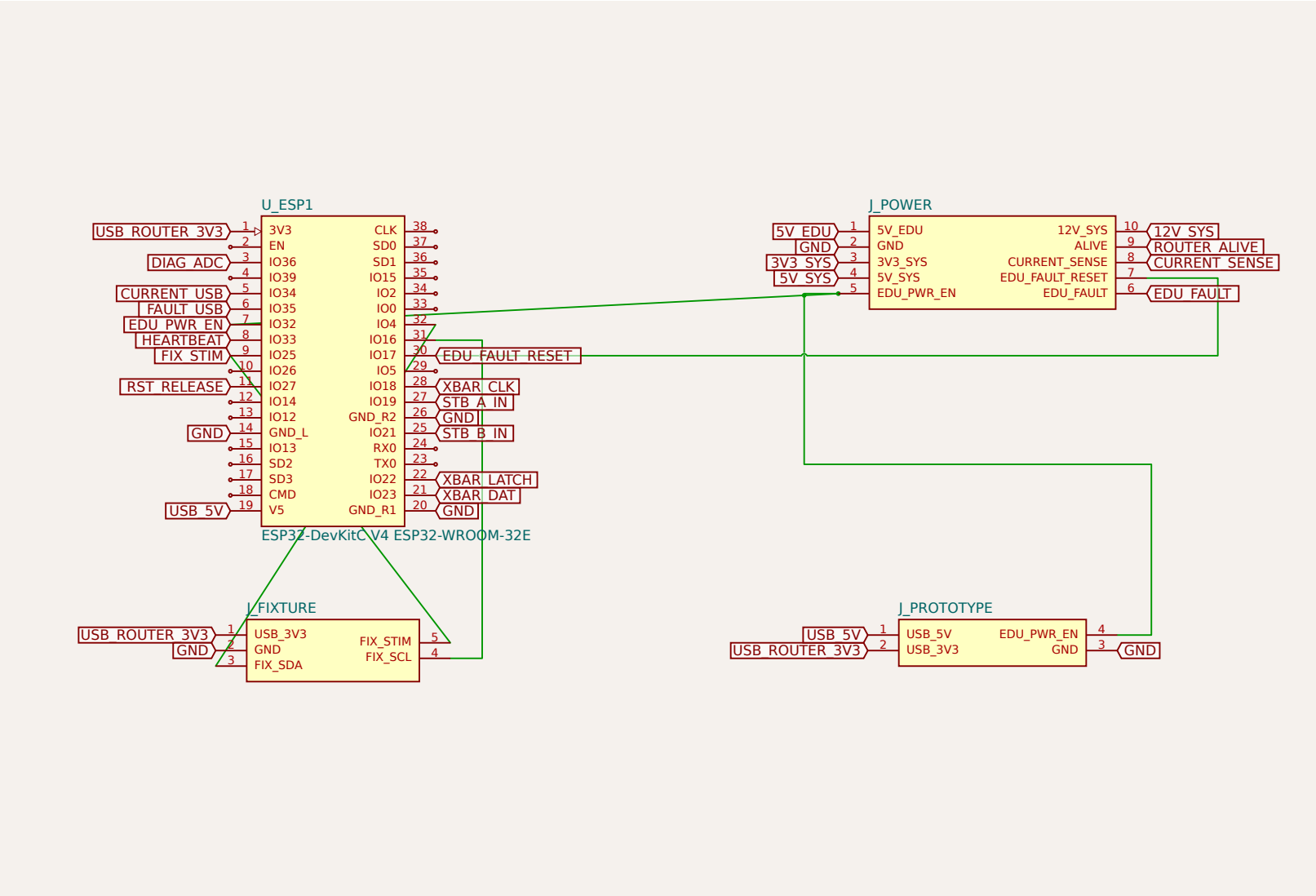
PROTOTYPE: physical fit and hardware tests pending.

router-core — numbered parts

Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

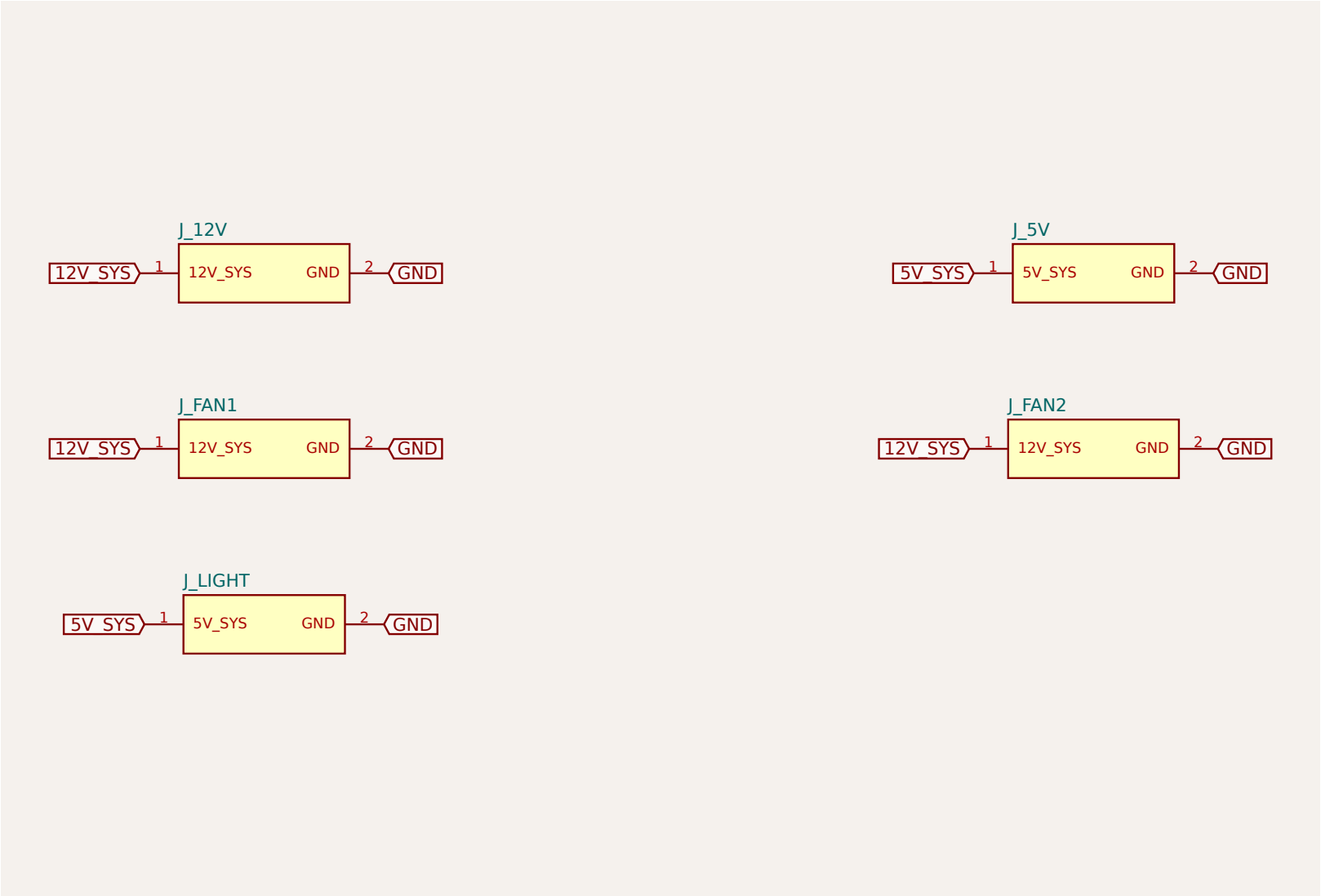
#	Reference	Purpose	Install	Adapter / carrier
1	U_ESP1	ESP32 DevKitC-38 — router	CP0	
2	J_POWER	POWER_MATRIX v2	CP11	
3	J_FIXTURE	manual continuity fixture; ESP1 GPIO4 SDA / GPIO16 SCL / GPIO25 stimulus	CP0	
4	J_PROTOTYPE	USB breadboard prototype; remove before final rail connection	CP0	
5	J_12V	Protected 12V input	CP32	
6	J_5V	System 5V input	CP32	
7	J_FAN1	12V fan 1 — fixed-speed	CP32	
8	J_FAN2	12V fan 2 — fixed-speed	CP32	
9	J_LIGHT	5V white lighting — never 5V_EDU	CP32	
10	J_LINK_1	Harness terminal: XBAR_DAT	CP0	
11	J_LINK_2	Harness terminal: XBAR_CLK	CP0	
12	J_LINK_3	Harness terminal: XBAR_LATCH	CP0	
13	J_LINK_4	Harness terminal: STB_A_IN	CP0	
14	J_LINK_5	Harness terminal: STB_B_IN	CP0	
15	J_LINK_6	Harness terminal: RST_RELEASE	CP0	
16	J_LINK_7	Harness terminal: HEARTBEAT	CP0	
17	J_LINK_8	Harness terminal: FAULT_USB	CP0	
18	J_LINK_9	Harness terminal: EDU_FAULT_RESET	CP0	
19	J_LINK_10	Harness terminal: CURRENT_USB	CP0	
20	J_LINK_11	Harness terminal: DIAG_ADC	CP0	

router-core — Controller



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

router-core — Circuit

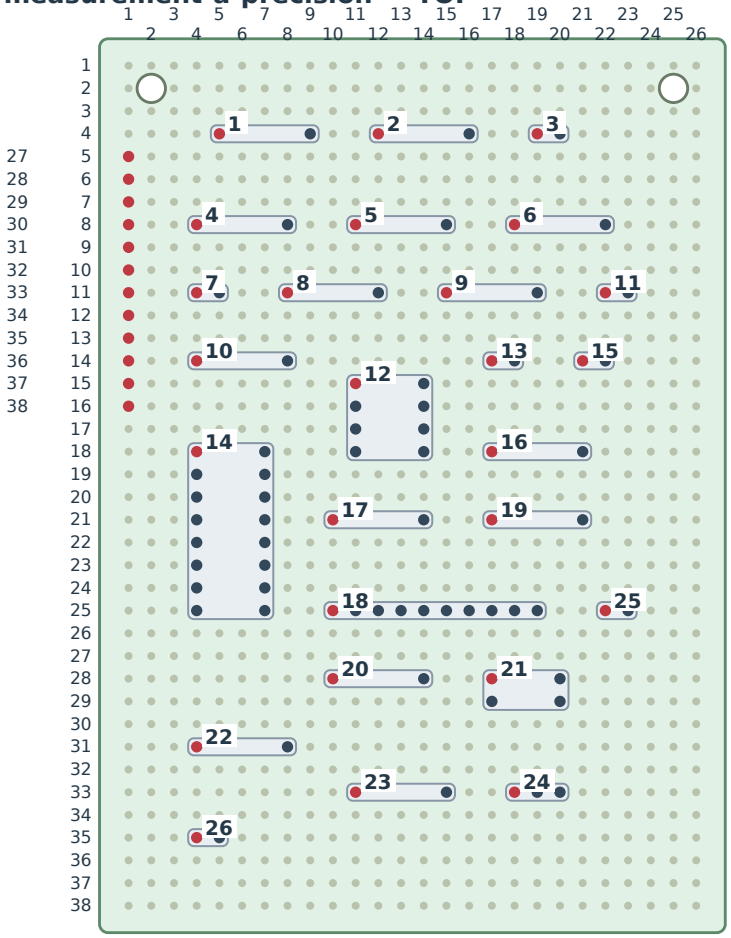


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-a-precision

Bank A: buffered voltage scanner and continuity stimulus

measurement-a-precision — TOP

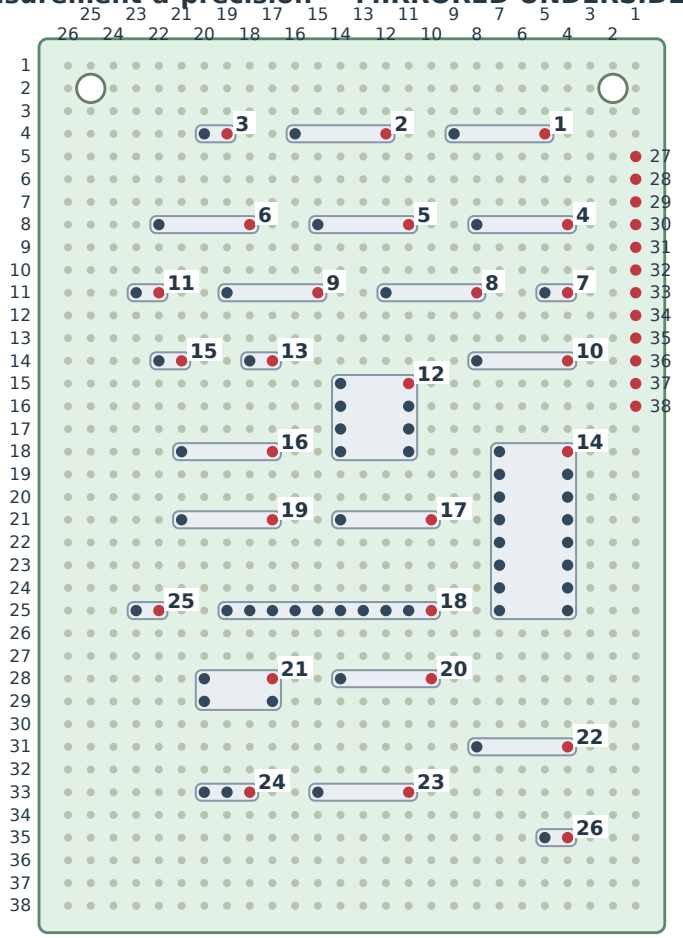


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

measurement-a-precision — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

measurement-a-precision — numbered parts

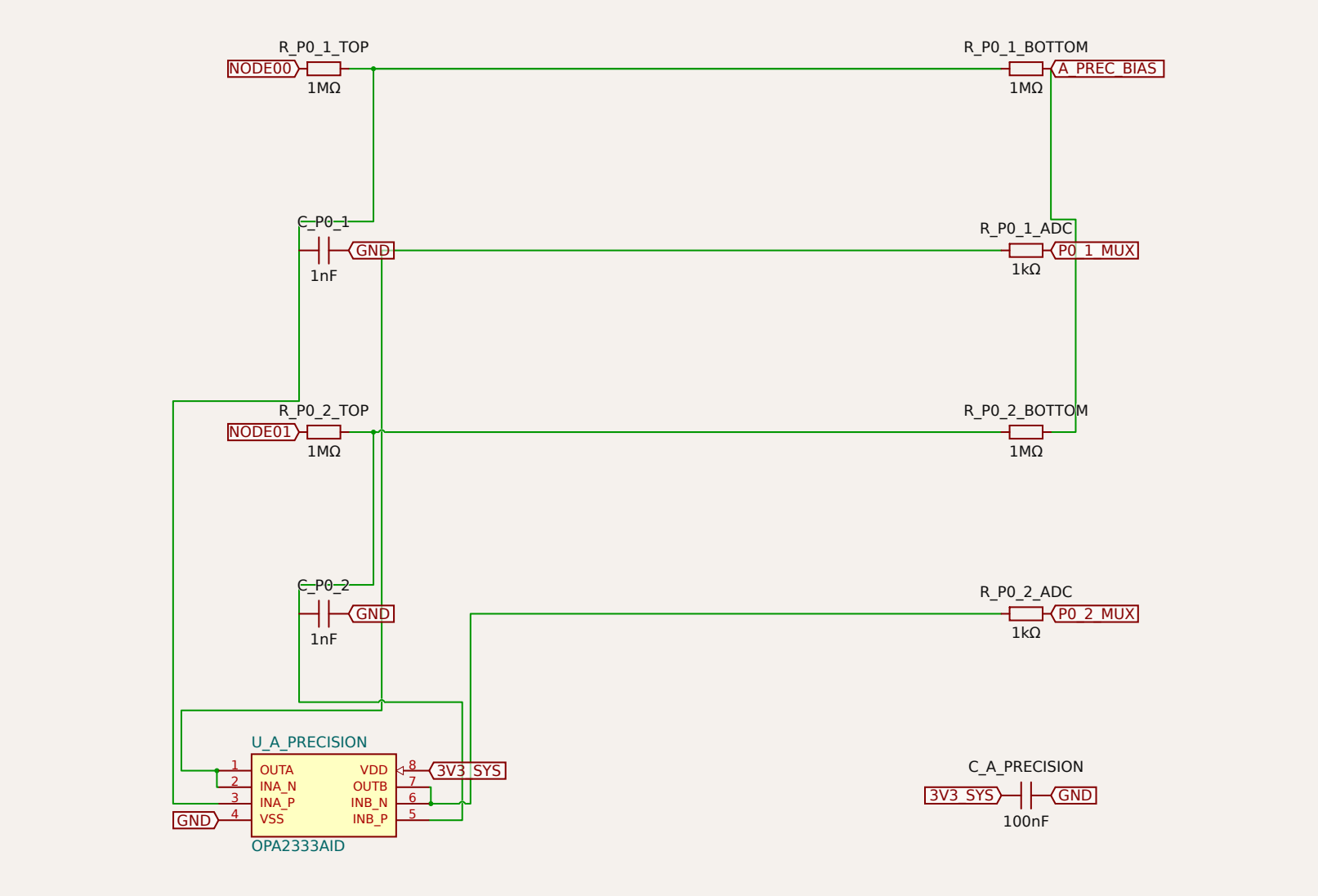
Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

#	Reference	Purpose	Install	Adapter / carrier
1	R_P0_1_TOP	R_P0_1_TOP: 1MΩ	CP1	
2	R_P0_1_BOTTOM	R_P0_1_BOTTOM: 1MΩ	CP1	
3	C_P0_1	C_P0_1: 1nFΩ	CP1	
4	R_P0_1_ADC	R_P0_1_ADC: 1kΩ	CP1	
5	R_P0_2_TOP	R_P0_2_TOP: 1MΩ	CP1	
6	R_P0_2_BOTTOM	R_P0_2_BOTTOM: 1MΩ	CP1	
7	C_P0_2	C_P0_2: 1nFΩ	CP1	
8	R_P0_2_ADC	R_P0_2_ADC: 1kΩ	CP1	
9	R_A_PREC_BIAS_TOP	R_A_PREC_BIAS_TOP: 5.62kΩ	CP1	
10	R_A_PREC_BIAS_BOTTOM	R_A_PREC_BIAS_BOTTOM: 1kΩ	CP1	
11	C_A_PREC_BIAS	C_A_PREC_BIAS: 100nFΩ	CP1	
12	U_A_PRECISION	OPA2333AID precision diagnostic/voltage buffer	CP1	SOIC8 to numbered DIP8
13	C_A_PRECISION	C_A_PRECISION: 100nFΩ	CP1	
14	U_A_MUX	CD4052BE buffered voltage selector	CP1	
15	C_A_MUX	C_A_MUX: 100nFΩ	CP1	
16	R_A_SEL_A	R_A_SEL_A: 100kΩ	CP1	
17	R_A_SEL_B	R_A_SEL_B: 100kΩ	CP1	
18	J_A_ADC_MODULE	ADS1115 carrier interface — verify labelled module pads	CP1	ADS1115 carrier
19	D_STIM	BAT85	CP1	
20	R_STIM	11kΩ ±1% independently bounds diagnostic stimulus below0.34mA	CP1	
21	K_STIM	AQY212EH	CP1	
22	R_LED	270Ω ±1% diagnostic PhotoMOS LED drive	CP1	
23	R_OFF	100k	CP1	
24	Q_STIM	2N7000	CP1	
25	J_DIAGNOSTIC	Removable diagnostic stimulus / return	CP1	
26	JP_RETURN	Fit only for unpowered continuity test	CP1	
27	J_LINK_1	Harness terminal: SCL_POSITION_0	CP1	
28	J_LINK_2	Harness terminal: SDA_POSITION_0	CP1	

measurement-a-precision — parts continued

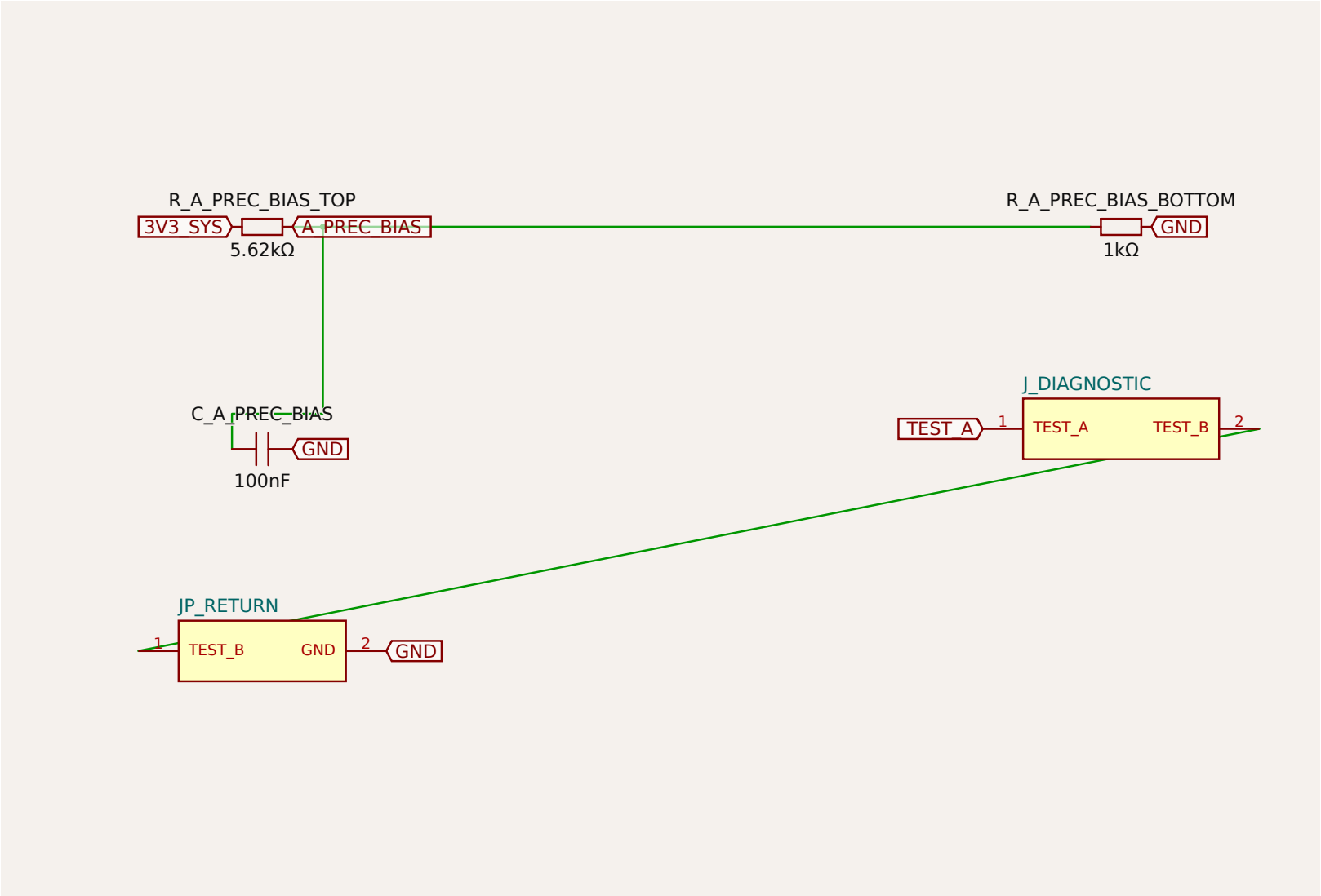
#	Reference	Purpose	Install	Adapter / carrier
29	J_LINK_3	Harness terminal: 3V3_SYS	CP1	
30	J_LINK_4	Harness terminal: NODE00	CP1	
31	J_LINK_5	Harness terminal: NODE01	CP1	
32	J_LINK_6	Harness terminal: FIX_STIM	CP1	
33	J_LINK_7	Harness terminal: MUX_A_A	CP1	
34	J_LINK_8	Harness terminal: MUX_A_B	CP1	
35	J_LINK_9	Harness terminal: P1_1_MUX	CP1	
36	J_LINK_10	Harness terminal: P1_2_MUX	CP1	
37	J_LINK_11	Harness terminal: P2_1_MUX	CP1	
38	J_LINK_12	Harness terminal: P2_2_MUX	CP1	

measurement-a-precision — Voltage inputs



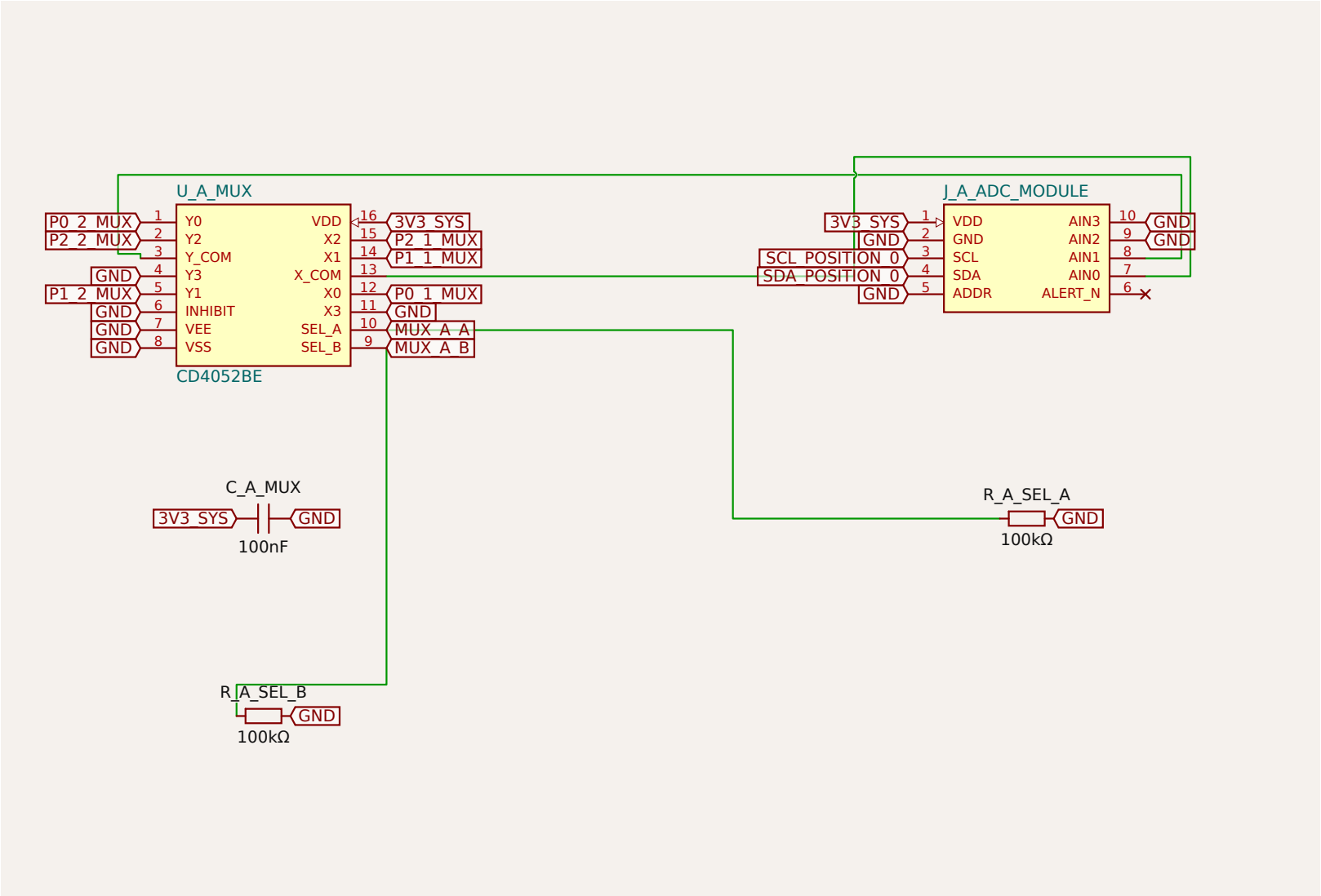
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-a-precision — Support



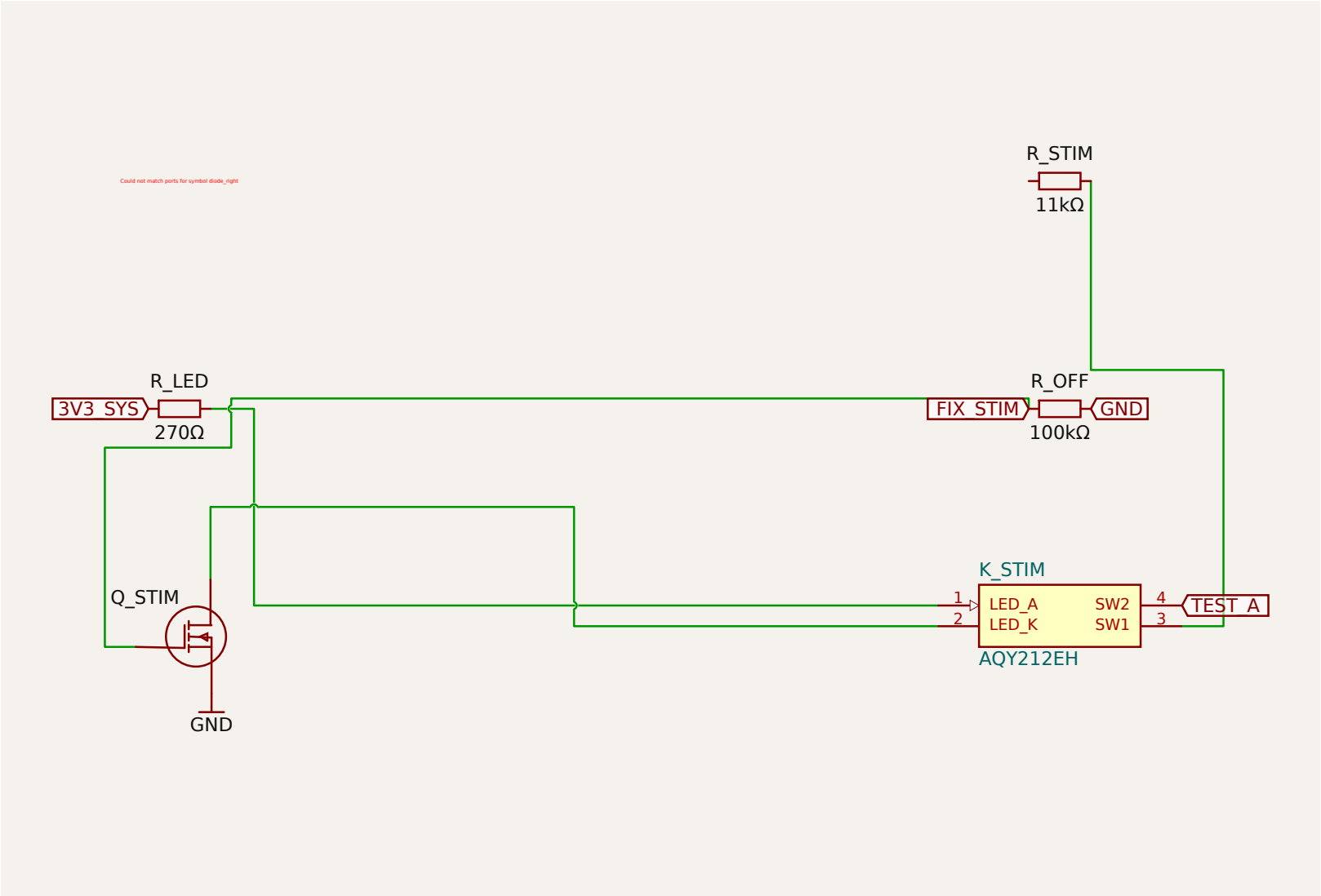
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-a-precision — Buffered selector and ADC



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-a-precision — Stimulus

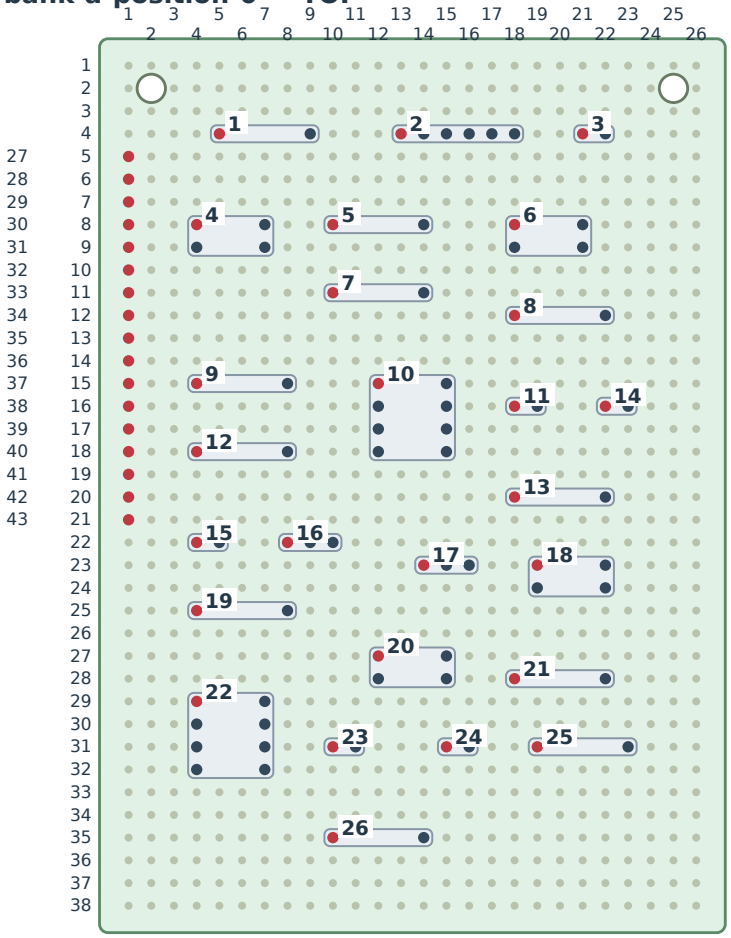


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-0

Bank A: position 0 complete isolated channel

bank-a-position-0 — TOP

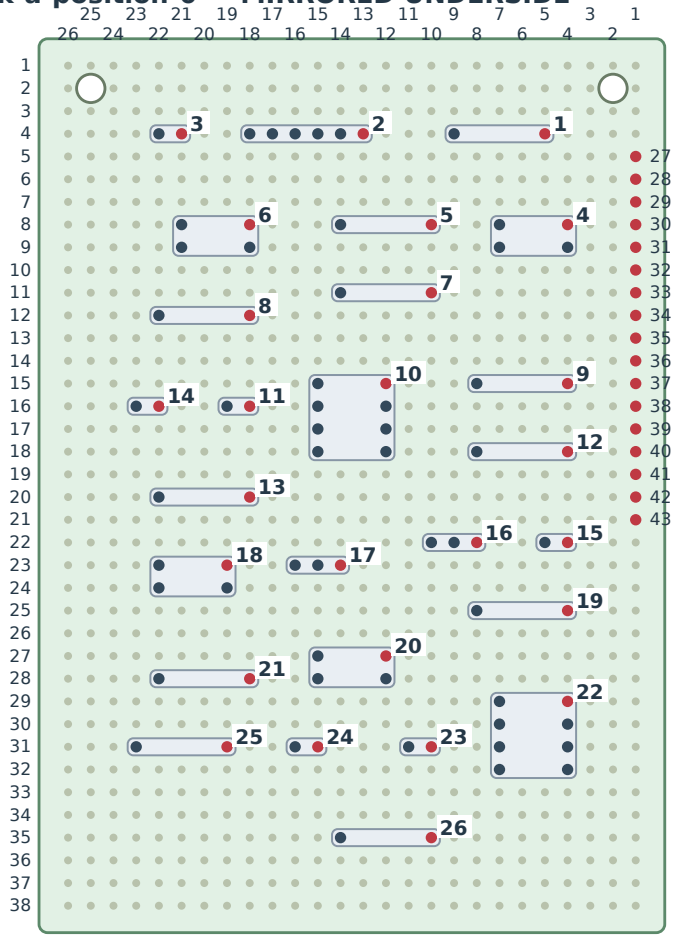


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

bank-a-position-0 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

bank-a-position-0 — numbered parts

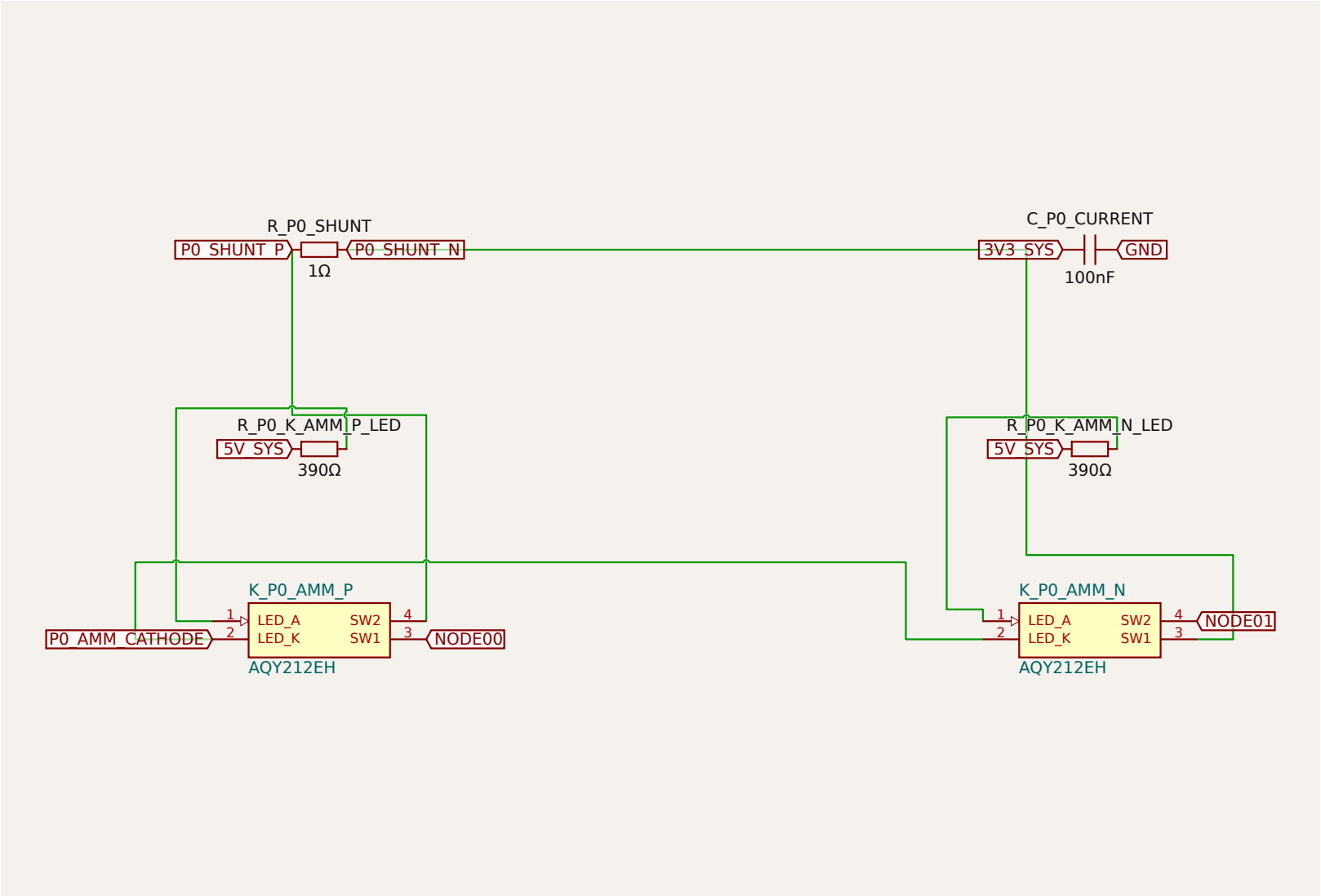
Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

#	Reference	Purpose	Install	Adapter / carrier
1	R_P0_SHUNT	1Ω ±1% external current shunt	CP22	
2	J_P0_CURRENT_MODULE	INA226 carrier: remove onboard shunt; external 1Ω	CP22	INA226 carrier
3	C_P0_CURRENT	100nF	CP22	
4	K_P0_AMM_P	AQY212EH	CP22	
5	R_P0_K_AMM_P_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP22	
6	K_P0_AMM_N	AQY212EH	CP22	
7	R_P0_K_AMM_N_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP22	
8	R_P0_SOURCE_SENSE	10	CP15	
9	R_P0_SINK_SENSE	10	CP15	
10	U_P0_LIMIT	MCP6002 dual source/sink clamp	CP15	
11	C_P0_LIMIT	100nF	CP15	
12	R_P0_HI_GATE	1k	CP15	
13	R_P0_LO_GATE	1k	CP15	
14	C_P0_HI_COMP	1nF	CP15	
15	C_P0_LO_COMP	1nF	CP15	
16	Q_P0_HIGH	BSS84	CP15	SOT23 to numbered 3-pin adapter
17	Q_P0_LOW	2N7000	CP15	
18	K_P0_HIGH	AQY212EH	CP15	
19	R_P0_K_HIGH_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP15	
20	K_P0_LOW	AQY212EH	CP15	
21	R_P0_K_LOW_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP15	
22	U_P0_BUS	TCA9517D	CP6	SOIC8 to DIP8
23	C_P0_BUS_A	100nF	CP6	
24	C_P0_BUS_B	100nF	CP6	
25	R_P0_SDA	4.7k	CP6	
26	R_P0_SCL	4.7k	CP6	
27	J_LINK_1	Harness terminal: SCL_POSITION_0	CP6	
28	J_LINK_2	Harness terminal: SDA_POSITION_0	CP6	

bank-a-position-0 — parts continued

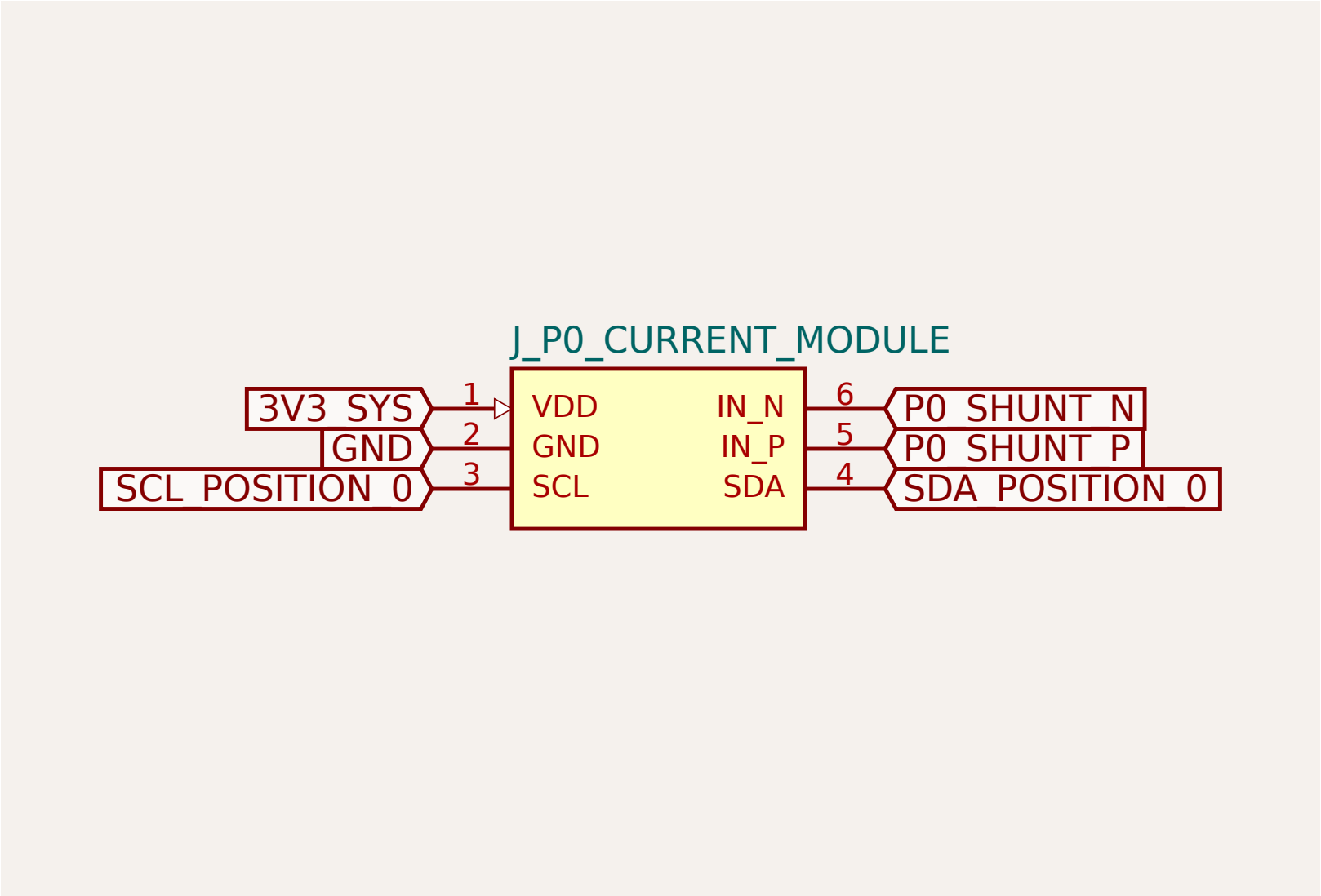
#	Reference	Purpose	Install	Adapter / carrier
29	J_LINK_3	Harness terminal: GND	CP6	
30	J_LINK_4	Harness terminal: 3V3_SYS	CP6	
31	J_LINK_5	Harness terminal: NODE00	CP22	
32	J_LINK_6	Harness terminal: NODE01	CP22	
33	J_LINK_7	Harness terminal: NODE02	CP15	
34	J_LINK_8	Harness terminal: 5V_SYS	CP15	
35	J_LINK_9	Harness terminal: 5V_EDU	CP15	
36	J_LINK_10	Harness terminal: USB_SMART_3V3	CP6	
37	J_LINK_11	Harness terminal: SCL_A	CP6	
38	J_LINK_12	Harness terminal: SDA_0	CP6	
39	J_LINK_13	Harness terminal: P0_K_HIGH_CATHODE	CP15	
40	J_LINK_14	Harness terminal: P0_K_LOW_CATHODE	CP15	
41	J_LINK_15	Harness terminal: P0_AMM_CATHODE	CP22	
42	J_LINK_16	Harness terminal: A_REF_HI	CP15	
43	J_LINK_17	Harness terminal: A_REF_LO	CP15	

bank-a-position-0 — Current



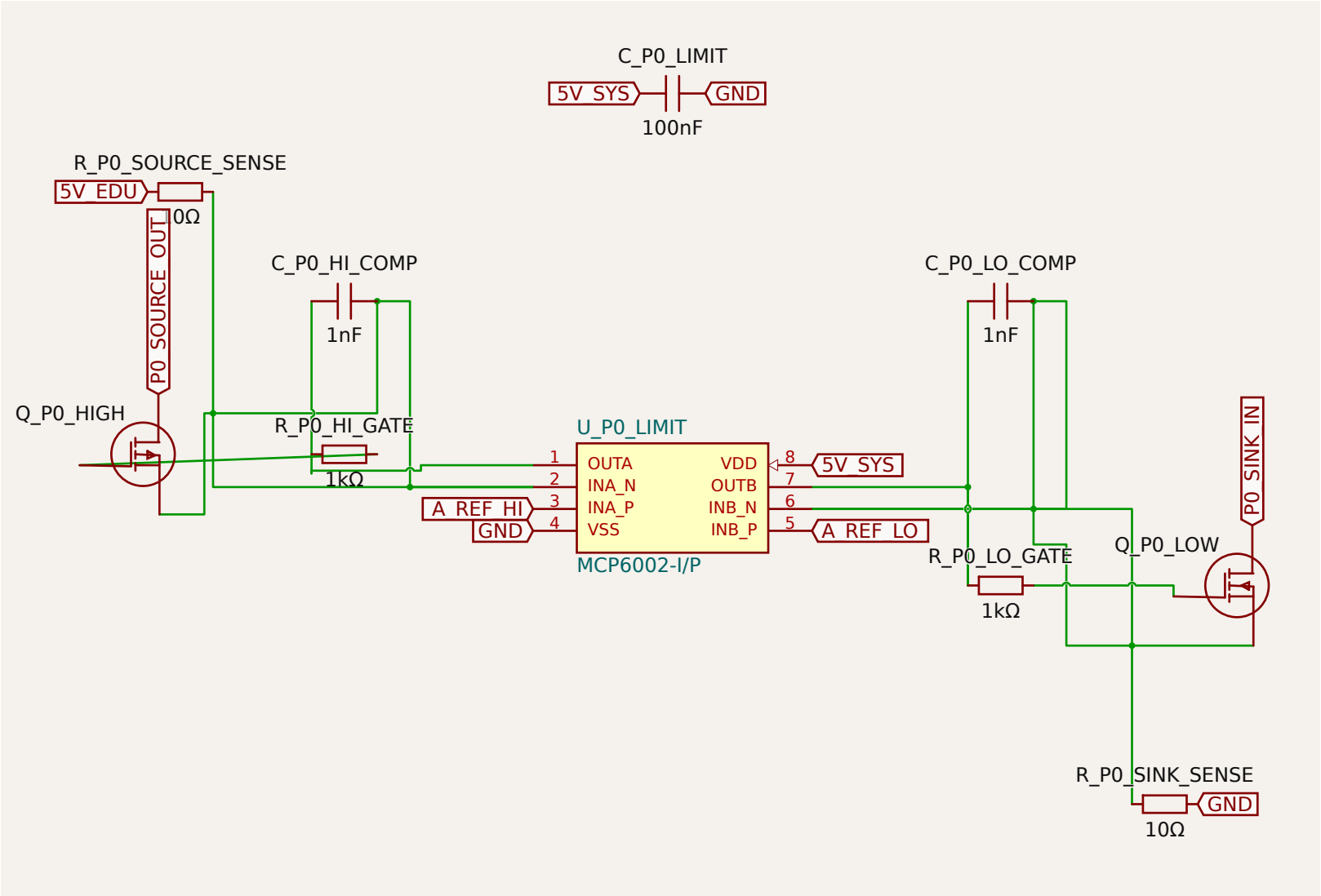
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-0 — Circuit



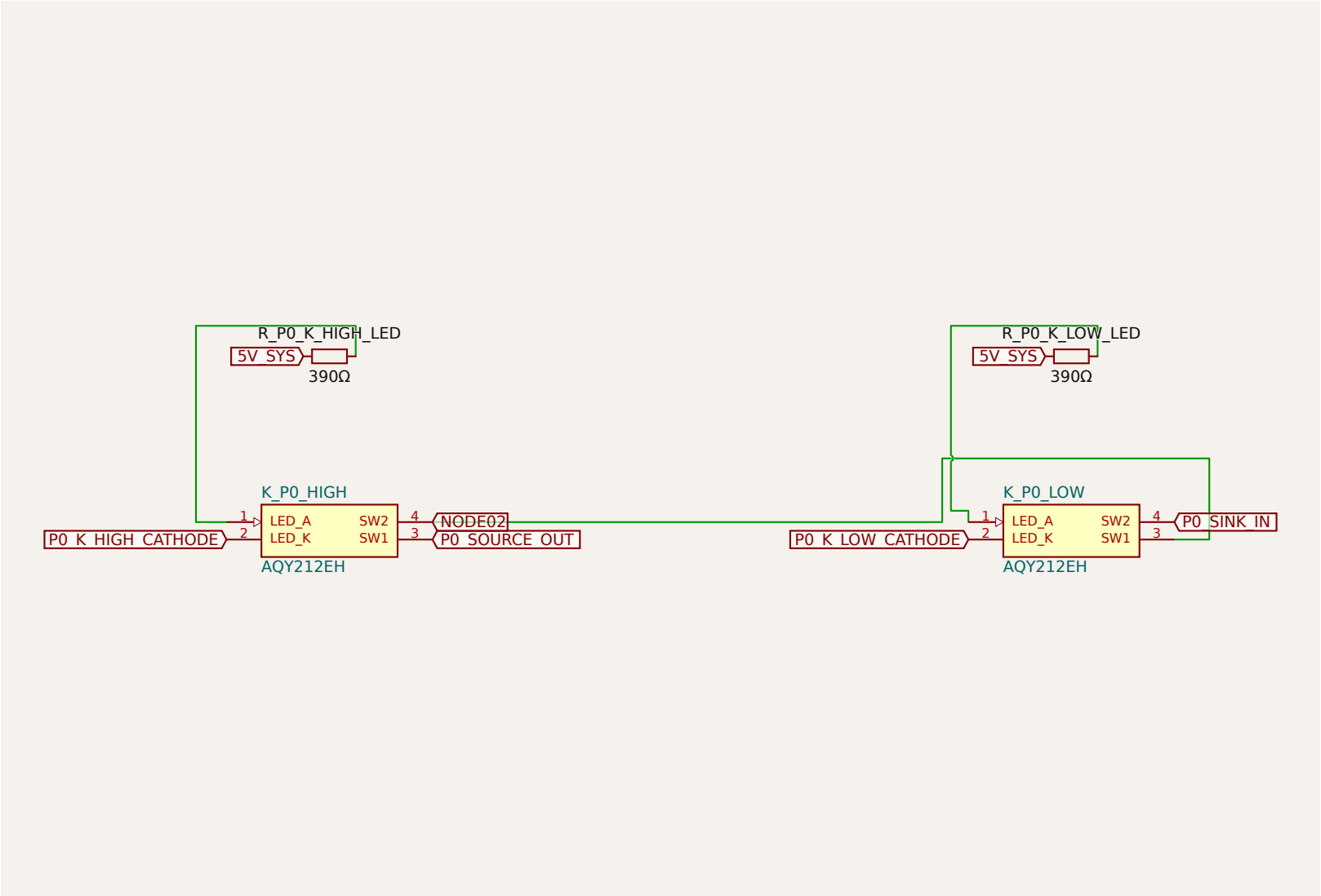
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-0 — Limits



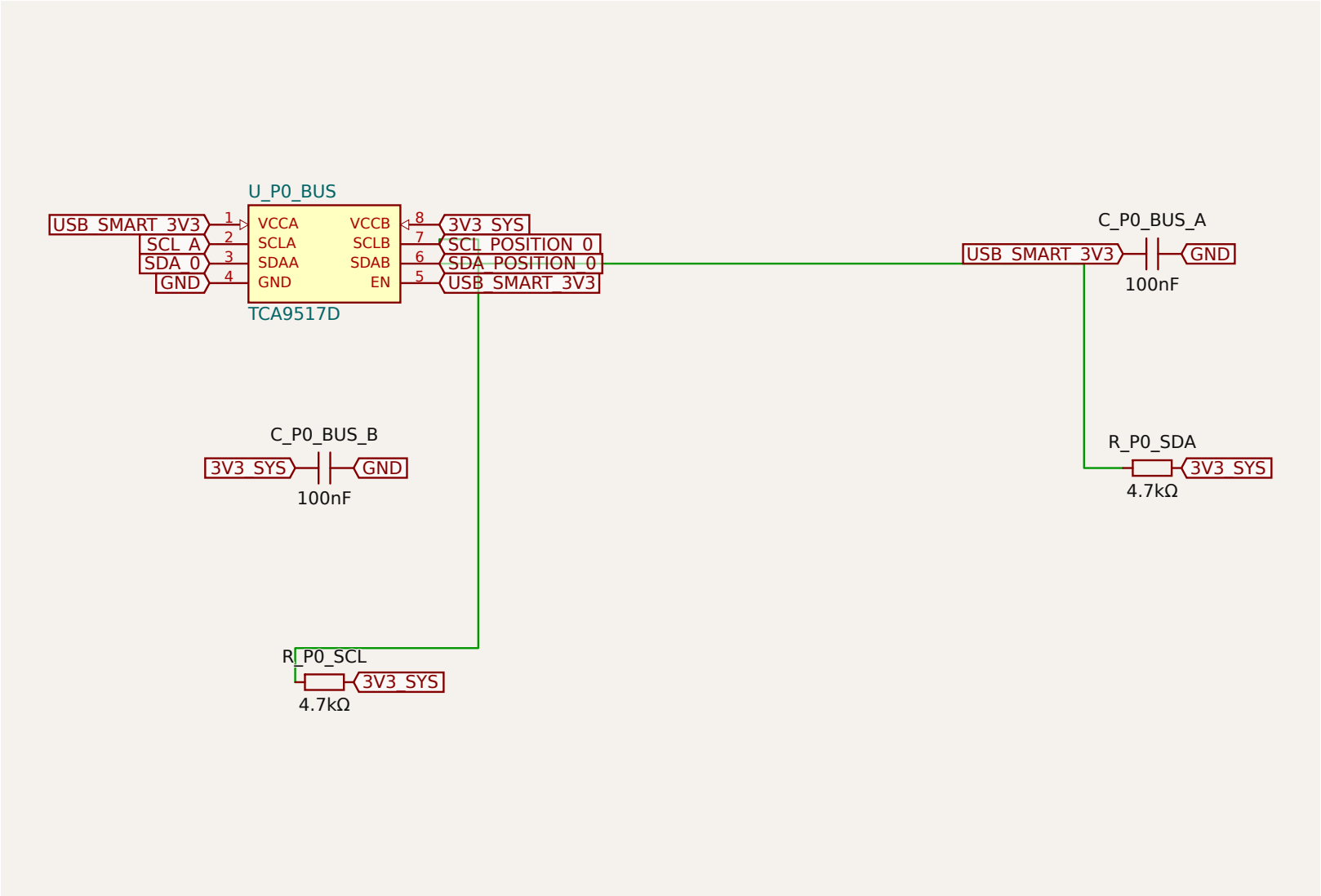
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-0 — Output



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-0 — Interface

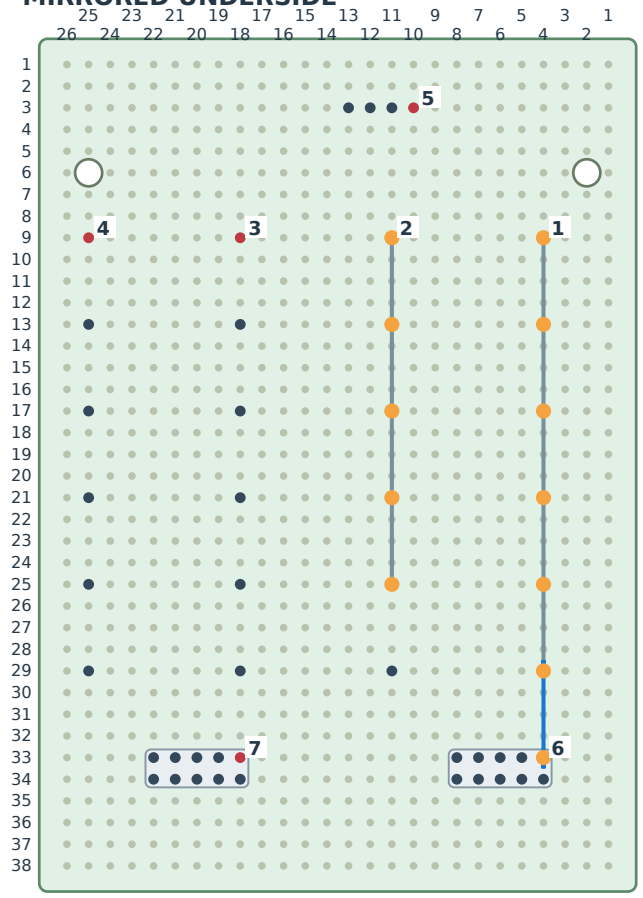


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

CP00 — a1

Install these wires after their endpoint components; inspect each net with power isolated.

a1 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

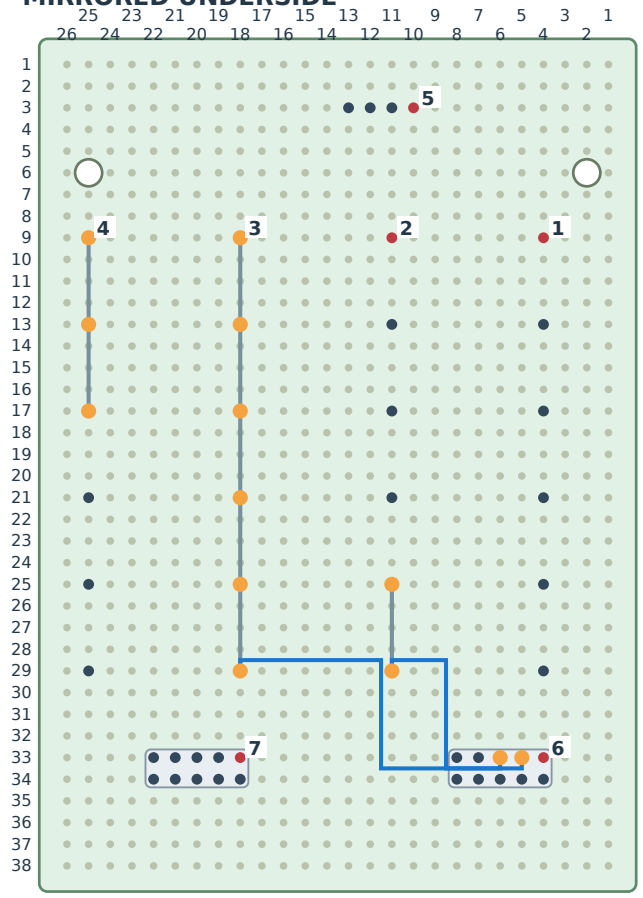
ID / net	From → to	TOP holes
W7 / NODE00	J_NODE00.1 → J_NODE00.2	(4,9) → (4,13)
W8 / NODE00	J_NODE00.2 → J_NODE00.3	(4,13) → (4,17)
W9 / NODE00	J_NODE00.3 → J_NODE00.4	(4,17) → (4,21)
W10 / NODE00	J_NODE00.4 → J_NODE00.5	(4,21) → (4,25)
W11 / NODE00	J_NODE00.5 → J_NODE00.6	(4,25) → (4,29)
W12 / NODE00	J_NODE00.6 → J_NODE.1	(4,29) → (4,33)
W14 / NODE01	J_NODE01.1 → J_NODE01.2	(11,9) → (11,13)
W15 / NODE01	J_NODE01.2 → J_NODE01.3	(11,13) → (11,17)
W16 / NODE01	J_NODE01.3 → J_NODE01.4	(11,17) → (11,21)
W17 / NODE01	J_NODE01.4 → J_NODE01.5	(11,21) → (11,25)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP00 — a1

Install these wires after their endpoint components; inspect each net with power isolated.

a1 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

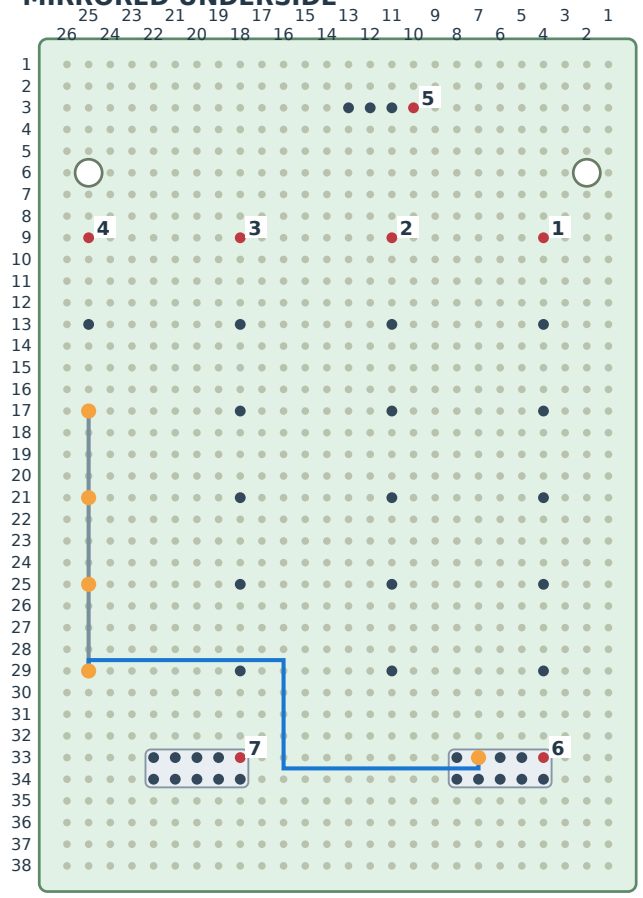
ID / net	From → to	TOP holes
W18 / NODE01	J_NODE01.5 → J_NODE01.6	(11,25) → (11,29)
W19 / NODE01	J_NODE01.6 → J_NODE.3	(11,29) → (5,33)
W21 / NODE02	J_NODE02.1 → J_NODE02.2	(18,9) → (18,13)
W22 / NODE02	J_NODE02.2 → J_NODE02.3	(18,13) → (18,17)
W23 / NODE02	J_NODE02.3 → J_NODE02.4	(18,17) → (18,21)
W24 / NODE02	J_NODE02.4 → J_NODE02.5	(18,21) → (18,25)
W25 / NODE02	J_NODE02.5 → J_NODE02.6	(18,25) → (18,29)
W26 / NODE02	J_NODE02.6 → J_NODE.5	(18,29) → (6,33)
W28 / NODE03	J_NODE03.1 → J_NODE03.2	(25,9) → (25,13)
W29 / NODE03	J_NODE03.2 → J_NODE03.3	(25,13) → (25,17)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP00 — a1

Install these wires after their endpoint components; inspect each net with power isolated.

a1 — MIRRORED UNDERSIDE



ID / net	From → to	TOP holes
W30 / NODE03	J_NODE03.3 → J_NODE03.4	(25,17) → (25,21)
W31 / NODE03	J_NODE03.4 → J_NODE03.5	(25,21) → (25,25)
W32 / NODE03	J_NODE03.5 → J_NODE03.6	(25,25) → (25,29)
W33 / NODE03	J_NODE03.6 → J_NODE.7	(25,29) → (7,33)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

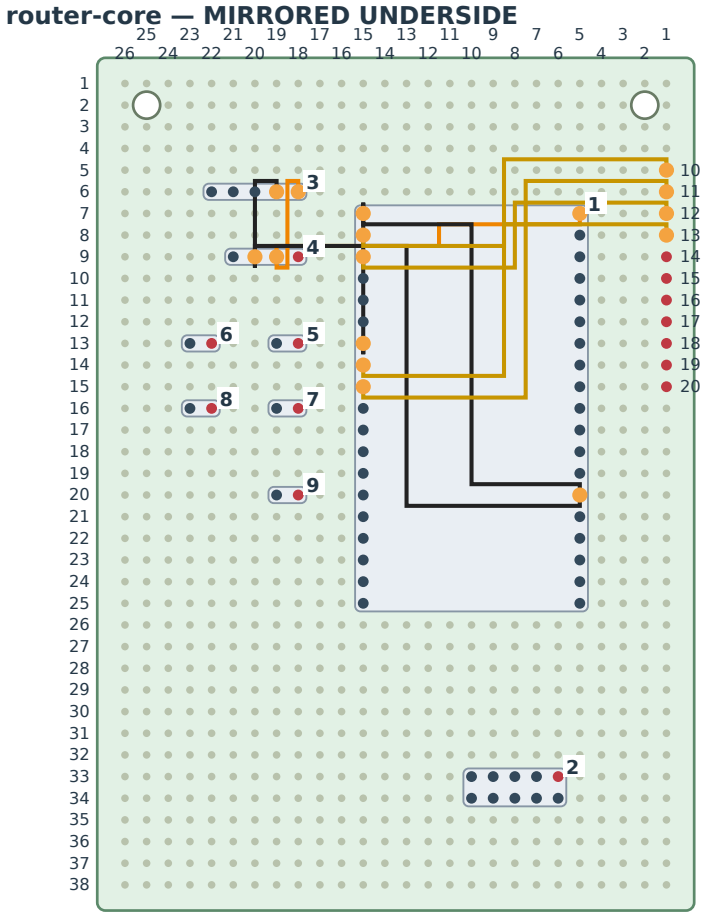
70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

CP00 — router-core

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

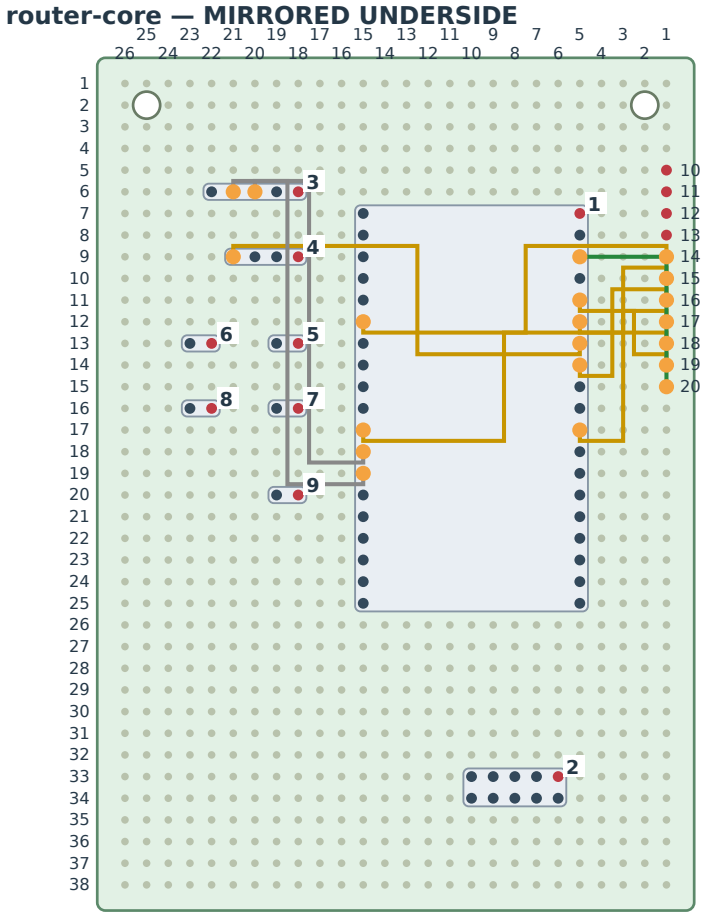
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / USB_ROUTER_3V3	J_FIXTURE.1 → J_PROTOTYPE.2	(18,6) → (19,9)
W2 / USB_ROUTER_3V3	J_PROTOTYPE.2 → U_ESP1.1	(19,9) → (5,7)
W3 / GND	J_FIXTURE.2 → J_PROTOTYPE.3	(19,6) → (20,9)
W4 / GND	J_PROTOTYPE.3 → U_ESP1.14	(20,9) → (5,20)
W5 / GND	U_ESP1.14 → U_ESP1.20	(5,20) → (15,7)
W6 / GND	U_ESP1.20 → U_ESP1.26	(15,7) → (15,13)
W13 / XBAR_DAT	J_LINK_1.1 → U_ESP1.21	(1,5) → (15,8)
W14 / XBAR_CLK	J_LINK_2.1 → U_ESP1.28	(1,6) → (15,15)
W15 / XBAR_LATCH	J_LINK_3.1 → U_ESP1.22	(1,7) → (15,9)
W16 / STB_A_IN	J_LINK_4.1 → U_ESP1.27	(1,8) → (15,14)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP00 — router-core

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

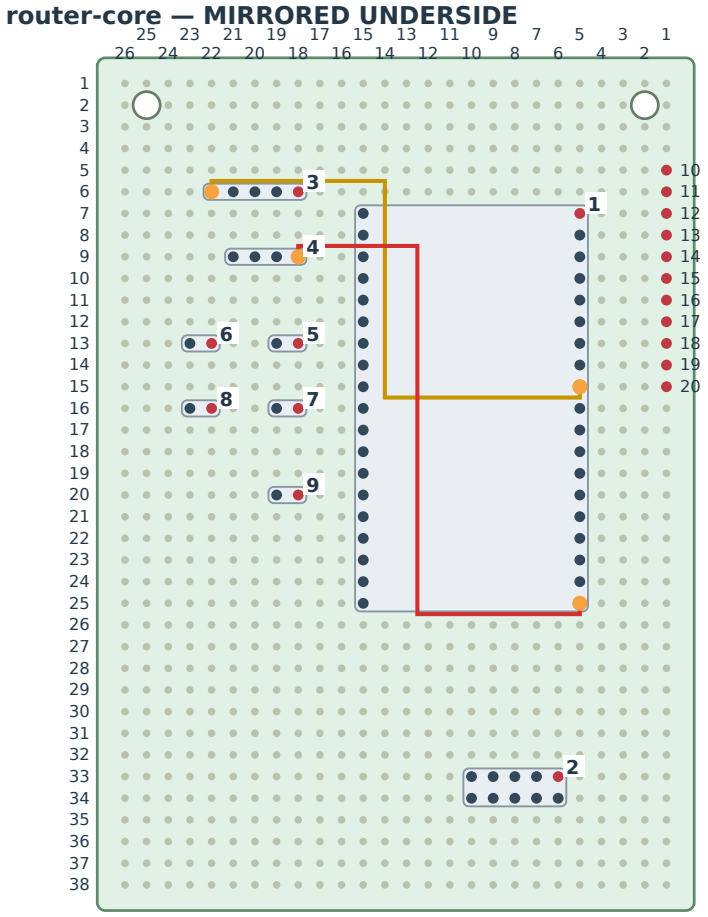
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W17 / STB_B_IN	J_LINK_5.1 → U_ESP1.25	(1,9) → (15,12)
W18 / RST_RELEASE	J_LINK_6.1 → U_ESP1.11	(1,10) → (5,17)
W19 / HEARTBEAT	J_LINK_7.1 → U_ESP1.8	(1,11) → (5,14)
W20 / EDU_PWR_EN	J_PROTOTYPE.4 → U_ESP1.7	(21,9) → (5,13)
W22 / FAULT_USB	J_LINK_8.1 → U_ESP1.6	(1,12) → (5,12)
W23 / EDU_FAULT_RESET	J_LINK_9.1 → U_ESP1.30	(1,13) → (15,17)
W25 / CURRENT_USB	J_LINK_10.1 → U_ESP1.5	(1,14) → (5,11)
W26 / DIAG_ADC	J_LINK_11.1 → U_ESP1.3	(1,15) → (5,9)
W27 / FIX_SDA	J_FIXTURE.3 → U_ESP1.32	(20,6) → (15,19)
W28 / FIX_SCL	J_FIXTURE.4 → U_ESP1.31	(21,6) → (15,18)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP00 — router-core

Install these wires after their endpoint components; inspect each net with power isolated.



ID / net	From → to	TOP holes
W29 / FIX_STIM	J_FIXTURE.5 → U_ESP1.9	(22,6) → (5,15)
W30 / USB_5V	J_PROTOTYPE.1 → U_ESP1.19	(18,9) → (5,25)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

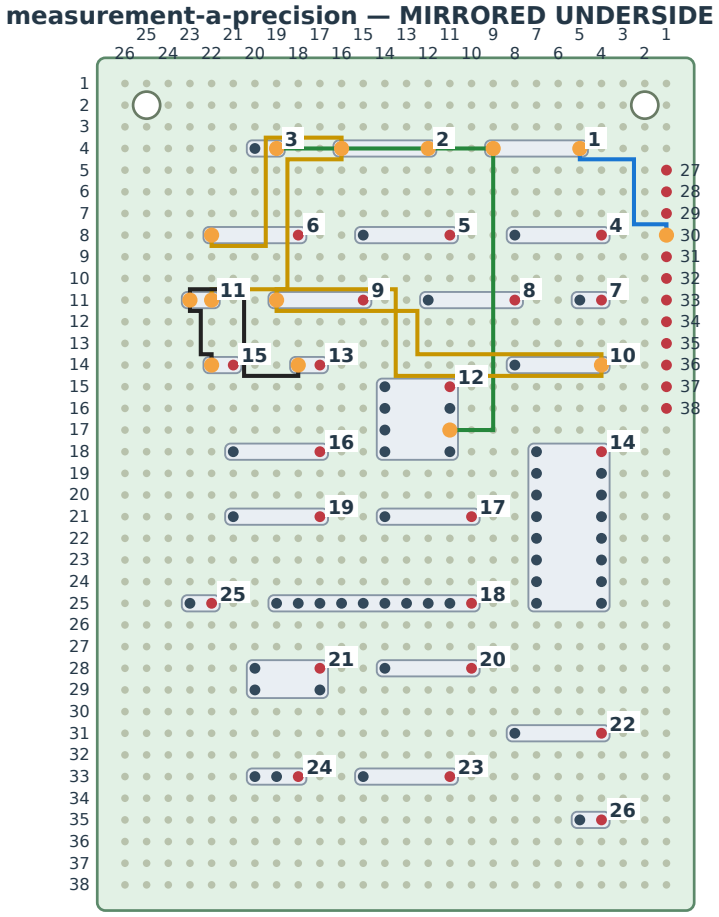
70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

CP01 — measurement-a-precision

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

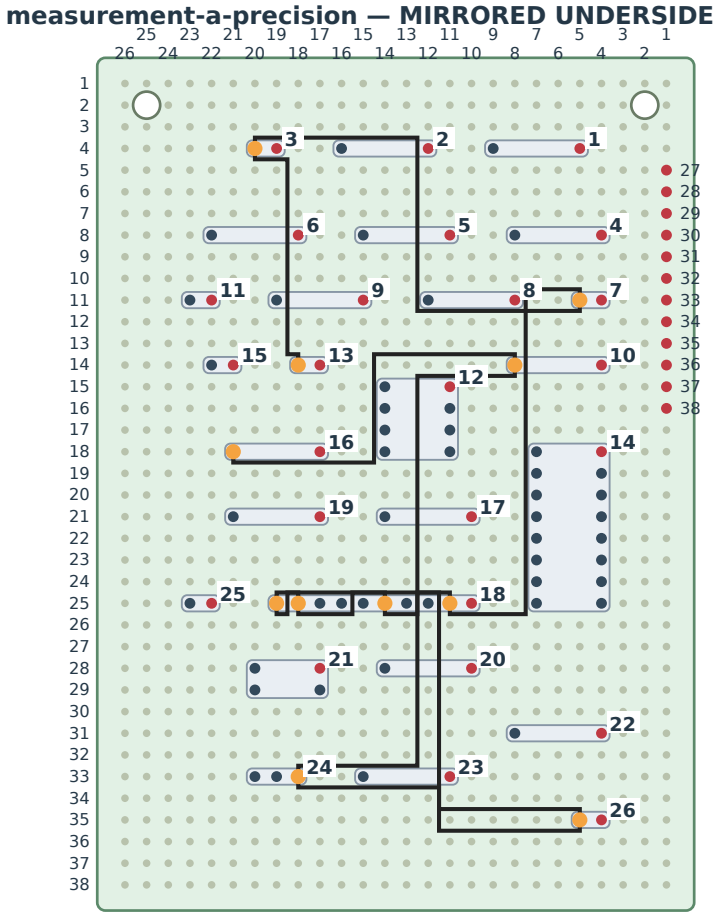
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / NODE00	J_LINK_4.1 → R_P0_1_TOP.1	(1,8) → (5,4)
W2 / P0_1_DIV	C_P0_1.1 → R_P0_1_BOTTOM.1	(19,4) → (12,4)
W3 / P0_1_DIV	R_P0_1_BOTTOM.1 → R_P0_1_TOP.2	(12,4) → (9,4)
W4 / P0_1_DIV	R_P0_1_TOP.2 → U_A_PRECISION.3	(9,4) → (11,17)
W5 / A_PREC_BIAS	C_A_PREC_BIAS.1 → R_A_PREC_BIAS_BOTTOM.1	(22,11) → (4,14)
W6 / A_PREC_BIAS	R_A_PREC_BIAS_BOTTOM.1 → R_A_PREC_BIAS_TOP.2	(4,14) → (19,11)
W7 / A_PREC_BIAS	R_A_PREC_BIAS_TOP.2 → R_P0_1_BOTTOM.2	(19,11) → (16,4)
W8 / A_PREC_BIAS	R_P0_1_BOTTOM.2 → R_P0_2_BOTTOM.2	(16,4) → (22,8)
W9 / GND	C_A_MUX.2 → C_A_PREC_BIAS.2	(22,14) → (23,11)
W10 / GND	C_A_PREC_BIAS.2 → C_A_PRECISION.2	(23,11) → (18,14)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP01 — measurement-a-precision

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

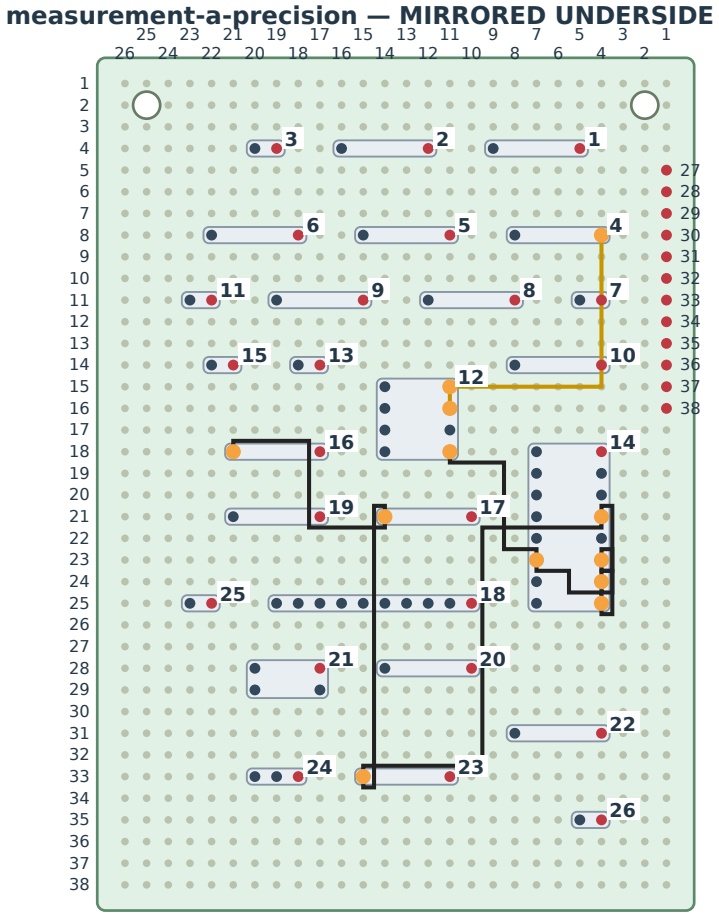
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W11 / GND	C_A_PRECISION.2 → C_P0_1.2	(18,14) → (20,4)
W12 / GND	C_P0_1.2 → C_P0_2.2	(20,4) → (5,11)
W13 / GND	C_P0_2.2 → J_A_ADC_MODULE.2	(5,11) → (11,25)
W14 / GND	J_A_ADC_MODULE.2 → J_A_ADC_MODULE.5	(11,25) → (14,25)
W15 / GND	J_A_ADC_MODULE.5 → J_A_ADC_MODULE.9	(14,25) → (18,25)
W16 / GND	J_A_ADC_MODULE.9 → J_A_ADC_MODULE.10	(18,25) → (19,25)
W17 / GND	J_A_ADC_MODULE.10 → JP_RETURN.2	(19,25) → (5,35)
W18 / GND	JP_RETURN.2 → Q_STIM.1	(5,35) → (18,33)
W19 / GND	Q_STIM.1 → R_A_PREC_BIAS_BOTTOM.2	(18,33) → (8,14)
W20 / GND	R_A_PREC_BIAS_BOTTOM.2 → R_A_SEL_A.2	(8,14) → (21,18)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP01 — measurement-a-precision

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

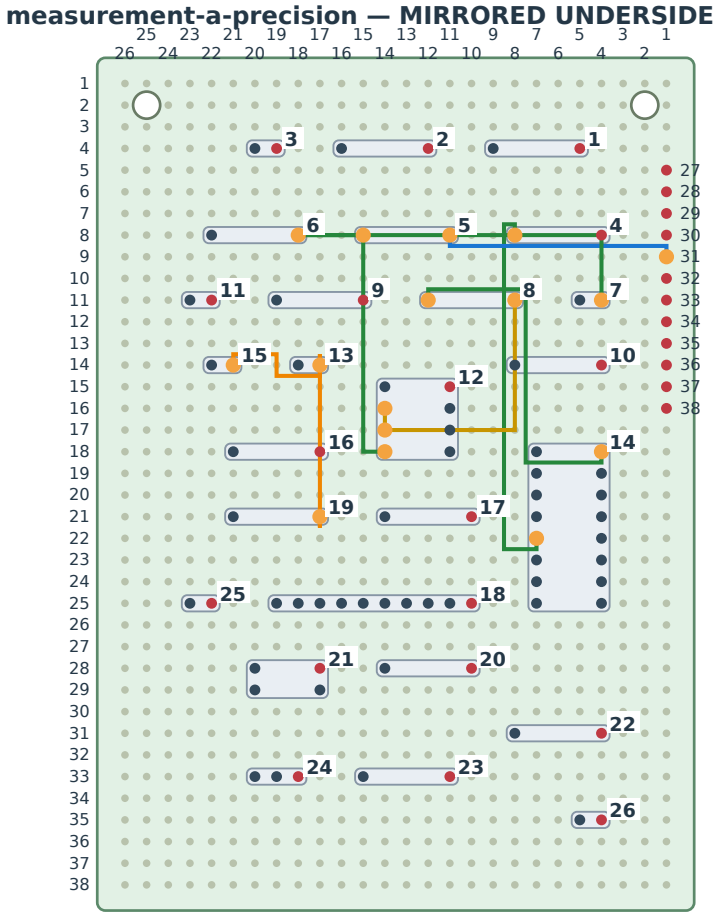
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W21 / GND	R_A_SEL_A.2 → R_A_SEL_B.2	(21,18) → (14,21)
W22 / GND	R_A_SEL_B.2 → R_OFF.2	(14,21) → (15,33)
W23 / GND	R_OFF.2 → U_A_MUX.4	(15,33) → (4,21)
W24 / GND	U_A_MUX.4 → U_A_MUX.6	(4,21) → (4,23)
W25 / GND	U_A_MUX.6 → U_A_MUX.7	(4,23) → (4,24)
W26 / GND	U_A_MUX.7 → U_A_MUX.8	(4,24) → (4,25)
W27 / GND	U_A_MUX.8 → U_A_MUX.11	(4,25) → (7,23)
W28 / GND	U_A_MUX.11 → U_A_PRECISION.4	(7,23) → (11,18)
W29 / P0_1_BUF	R_P0_1_ADC.1 → U_A_PRECISION.1	(4,8) → (11,15)
W30 / P0_1_BUF	U_A_PRECISION.1 → U_A_PRECISION.2	(11,15) → (11,16)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP01 — measurement-a-precision

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

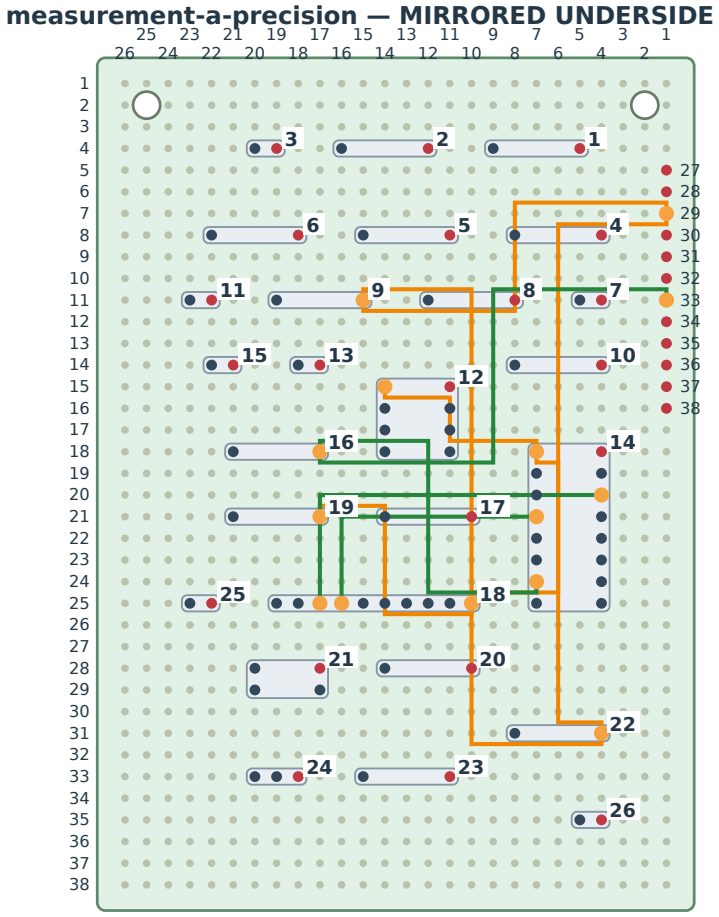
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W31 / P0_1_MUX	R_P0_1_ADC.2 → U_A_MUX.12	(8,8) → (7,22)
W32 / NODE01	J_LINK_5.1 → R_P0_2_TOP.1	(1,9) → (11,8)
W33 / P0_2_DIV	C_P0_2.1 → R_P0_2_BOTTOM.1	(4,11) → (18,8)
W34 / P0_2_DIV	R_P0_2_BOTTOM.1 → R_P0_2_TOP.2	(18,8) → (15,8)
W35 / P0_2_DIV	R_P0_2_TOP.2 → U_A_PRECISION.5	(15,8) → (14,18)
W36 / P0_2_BUF	R_P0_2_ADC.1 → U_A_PRECISION.6	(8,11) → (14,17)
W37 / P0_2_BUF	U_A_PRECISION.6 → U_A_PRECISION.7	(14,17) → (14,16)
W38 / P0_2_MUX	R_P0_2_ADC.2 → U_A_MUX.1	(12,11) → (4,18)
W39 / 3V3_SYS	C_A_MUX.1 → C_A_PRECISION.1	(21,14) → (17,14)
W40 / 3V3_SYS	C_A_PRECISION.1 → D_STIM.1	(17,14) → (17,21)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP01 — measurement-a-precision

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

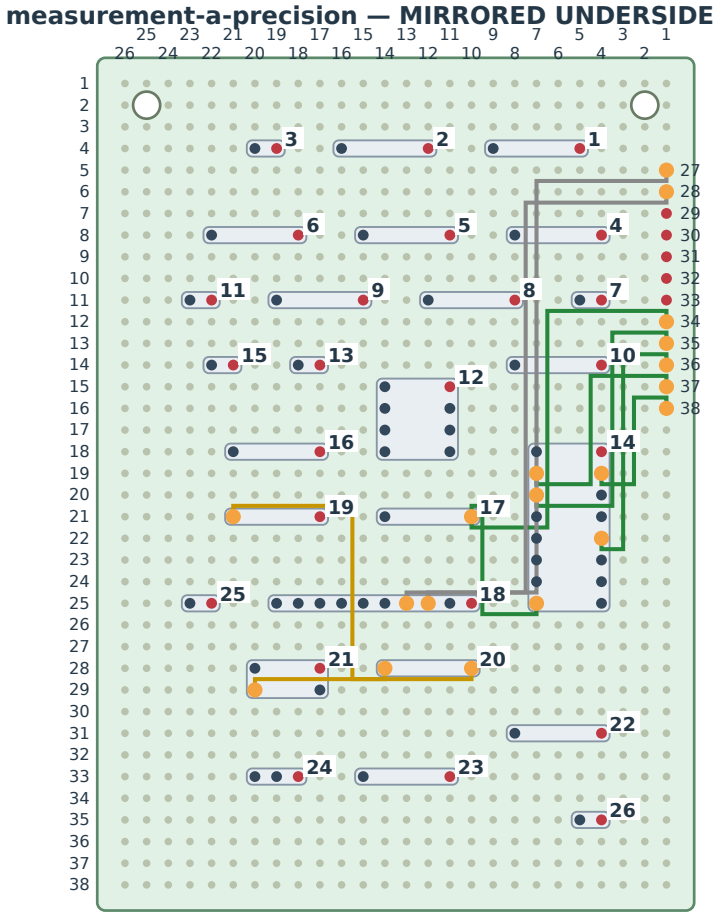
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W41 / 3V3_SYS	D_STIM.1 → J_A_ADC_MODULE.1	(17,21) → (10,25)
W42 / 3V3_SYS	J_A_ADC_MODULE.1 → J_LINK_3.1	(10,25) → (1,7)
W43 / 3V3_SYS	J_LINK_3.1 → R_A_PREC_BIAS_TOP.1	(1,7) → (15,11)
W44 / 3V3_SYS	R_A_PREC_BIAS_TOP.1 → R_LED.1	(15,11) → (4,31)
W45 / 3V3_SYS	R_LED.1 → U_A_MUX.16	(4,31) → (7,18)
W46 / 3V3_SYS	U_A_MUX.16 → U_A_PRECISION.8	(7,18) → (14,15)
W47 / ADC_A_P2	J_A_ADC_MODULE.8 → U_A_MUX.3	(17,25) → (4,20)
W48 / ADC_A_P1	J_A_ADC_MODULE.7 → U_A_MUX.13	(16,25) → (7,21)
W49 / MUX_A_A	J_LINK_7.1 → R_A_SEL_A.1	(1,11) → (17,18)
W50 / MUX_A_A	R_A_SEL_A.1 → U_A_MUX.10	(17,18) → (7,24)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP01 — measurement-a-precision

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

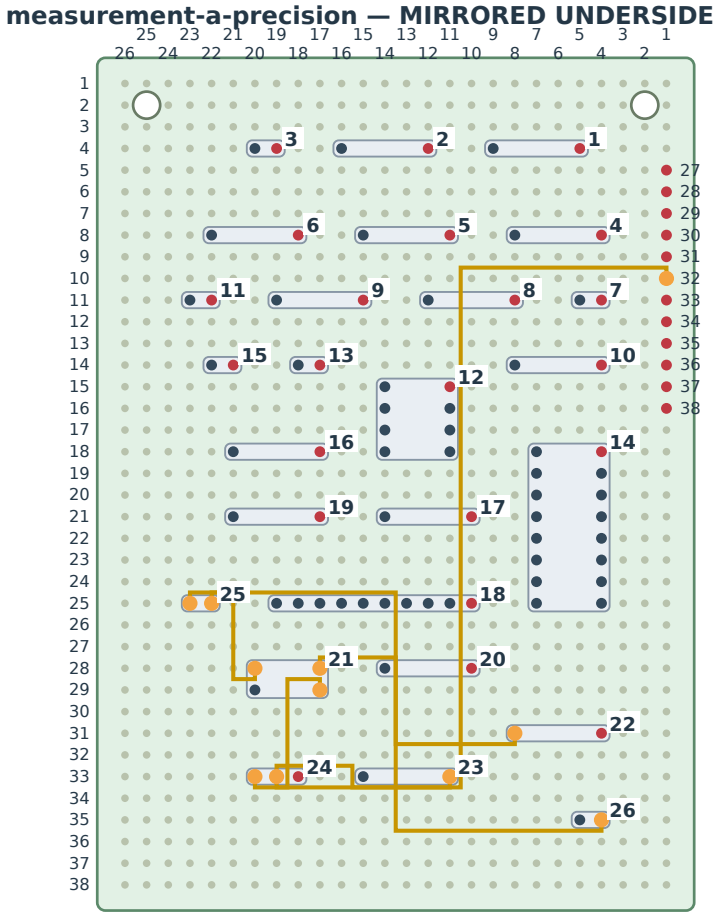
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W51 / MUX_A_B	J_LINK_8.1 → R_A_SEL_B.1	(1,12) → (10,21)
W52 / MUX_A_B	R_A_SEL_B.1 → U_A_MUX.9	(10,21) → (7,25)
W53 / P1_1_MUX	J_LINK_9.1 → U_A_MUX.14	(1,13) → (7,20)
W54 / P1_2_MUX	J_LINK_10.1 → U_A_MUX.5	(1,14) → (4,22)
W55 / P2_1_MUX	J_LINK_11.1 → U_A_MUX.15	(1,15) → (7,19)
W56 / P2_2_MUX	J_LINK_12.1 → U_A_MUX.2	(1,16) → (4,19)
W57 / SCL_POSITION_0	J_A_ADC_MODULE.3 → J_LINK_1.1	(12,25) → (1,5)
W58 / SDA_POSITION_0	J_A_ADC_MODULE.4 → J_LINK_2.1	(13,25) → (1,6)
W59 / STIM_FEED	D_STIM.2 → R_STIM.1	(21,21) → (10,28)
W60 / STIM_LIMIT	K_STIM.3 → R_STIM.2	(20,29) → (14,28)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP01 — measurement-a-precision

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

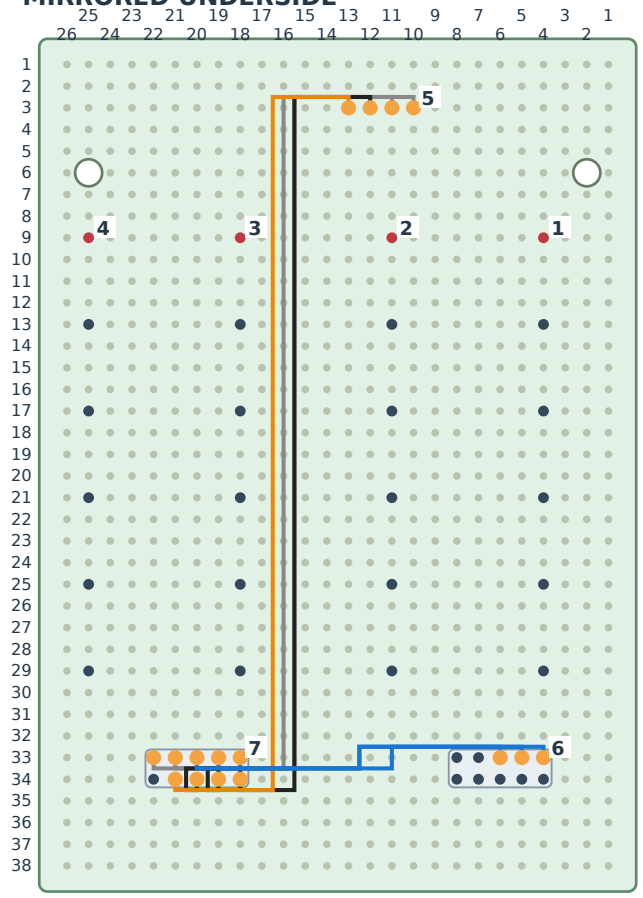
ID / net	From → to	TOP holes
W61 / STIM_LED_A	K_STIM.1 → R_LED.2	(17,28) → (8,31)
W62 / STIM_LED_K	K_STIM.2 → Q_STIM.3	(17,29) → (20,33)
W63 / TEST_A	J_DIAGNOSTIC.1 → K_STIM.4	(22,25) → (20,28)
W64 / FIX_STIM	J_LINK_6.1 → Q_STIM.2	(1,10) → (19,33)
W65 / FIX_STIM	Q_STIM.2 → R_OFF.1	(19,33) → (11,33)
W66 / TEST_B	J_DIAGNOSTIC.2 → JP_RETURN.1	(23,25) → (4,35)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP06 — a1

Install these wires after their endpoint components; inspect each net with power isolated.

a1 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

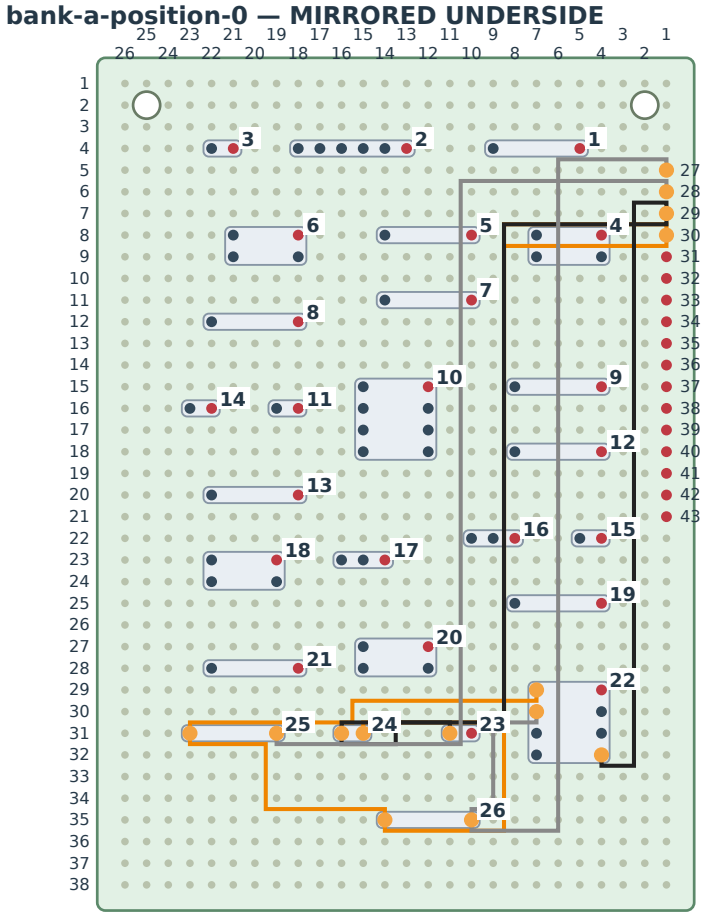
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / SCL_POSITION_0	J_SERVICE.1 → J_SMART.9	(10,3) → (22,33)
W2 / SDA_POSITION_0	J_SERVICE.2 → J_SMART.7	(11,3) → (21,33)
W3 / GND	J_SERVICE.3 → J_SMART.2	(12,3) → (18,34)
W4 / GND	J_SMART.2 → J_SMART.4	(18,34) → (19,34)
W5 / GND	J_SMART.4 → J_SMART.6	(19,34) → (20,34)
W6 / 3V3_SYS	J_SERVICE.4 → J_SMART.8	(13,3) → (21,34)
W13 / NODE00	J_NODE.1 → J_SMART.1	(4,33) → (18,33)
W20 / NODE01	J_NODE.3 → J_SMART.3	(5,33) → (19,33)
W27 / NODE02	J_NODE.5 → J_SMART.5	(6,33) → (20,33)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP06 — bank-a-position-0

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

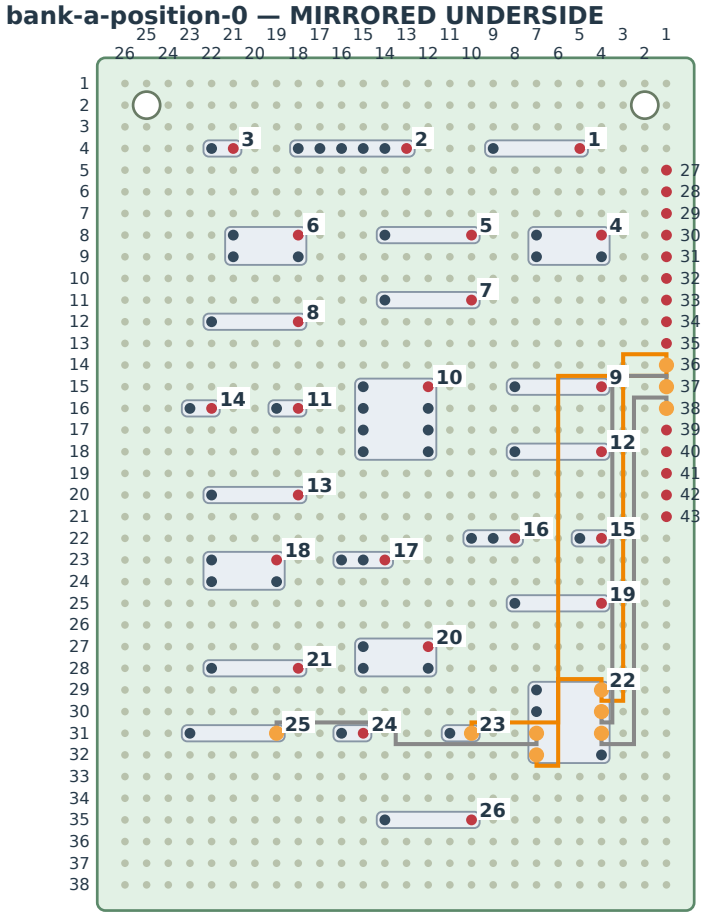
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W5 / 3V3_SYS	C_P0_BUS_B.1 → J_LINK_4.1	(15,31) → (1,8)
W6 / 3V3_SYS	J_LINK_4.1 → R_P0_SCL.2	(1,8) → (14,35)
W7 / 3V3_SYS	R_P0_SCL.2 → R_P0_SDA.2	(14,35) → (23,31)
W8 / 3V3_SYS	R_P0_SDA.2 → U_P0_BUS.8	(23,31) → (7,29)
W11 / GND	C_P0_BUS_A.2 → C_P0_BUS_B.2	(11,31) → (16,31)
W12 / GND	C_P0_BUS_B.2 → J_LINK_3.1	(16,31) → (1,7)
W13 / GND	J_LINK_3.1 → U_P0_BUS.4	(1,7) → (4,32)
W19 / SCL_POSITION_0	J_LINK_1.1 → R_P0_SCL.1	(1,5) → (10,35)
W20 / SCL_POSITION_0	R_P0_SCL.1 → U_P0_BUS.7	(10,35) → (7,30)
W22 / SDA_POSITION_0	J_LINK_2.1 → R_P0_SDA.1	(1,6) → (19,31)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP06 — bank-a-position-0

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W23 / SDA_POSITION_0	R_P0_SDA.1 → U_P0_BUS.6	(19,31) → (7,31)
W60 / USB_SMART_3V3	C_P0_BUS_A.1 → J_LINK_10.1	(10,31) → (1,14)
W61 / USB_SMART_3V3	J_LINK_10.1 → U_P0_BUS.1	(1,14) → (4,29)
W62 / USB_SMART_3V3	U_P0_BUS.1 → U_P0_BUS.5	(4,29) → (7,32)
W63 / SCL_A	J_LINK_11.1 → U_P0_BUS.2	(1,15) → (4,30)
W64 / SDA_0	J_LINK_12.1 → U_P0_BUS.3	(1,16) → (4,31)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP01 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H5 / GND	router-core / J_FIXTURE.2 TOP (19,6)	measurement-a-precision / JP_RETURN.2 TOP (5,35)
H155 / FIX_STIM	router-core / J_FIXTURE.5 TOP (22,6)	measurement-a-precision / J_LINK_6.1 TOP (1,10)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP06 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H1 / SCL_POSITION_0	measurement-a-precision / J_LINK_1.1 TOP (1,5)	a1 / J_SMART.9 TOP (22,33)
H2 / SCL_POSITION_0	a1 / J_SMART.9 TOP (22,33)	bank-a-position-0 / J_LINK_1.1 TOP (1,5)
H3 / SDA_POSITION_0	measurement-a-precision / J_LINK_2.1 TOP (1,6)	a1 / J_SMART.7 TOP (21,33)
H4 / SDA_POSITION_0	a1 / J_SMART.7 TOP (21,33)	bank-a-position-0 / J_LINK_2.1 TOP (1,6)
H6 / GND	measurement-a-precision / JP_RETURN.2 TOP (5,35)	a1 / J_SMART.2 TOP (18,34)
H7 / GND	a1 / J_SMART.2 TOP (18,34)	bank-a-position-0 / J_LINK_3.1 TOP (1,7)
H31 / 3V3_SYS	measurement-a-precision / J_LINK_3.1 TOP (1,7)	a1 / J_SMART.8 TOP (21,34)
H32 / 3V3_SYS	a1 / J_SMART.8 TOP (21,34)	bank-a-position-0 / J_LINK_4.1 TOP (1,8)
H52 / NODE00	a1 / J_NODE.1 TOP (4,33)	measurement-a-precision / J_LINK_4.1 TOP (1,8)
H55 / NODE01	a1 / J_NODE.3 TOP (5,33)	measurement-a-precision / J_LINK_5.1 TOP (1,9)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.